

Principles of Quantum Mechanics – R. Shankar

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1 Mathematical Introduction

1.1 The Delta function

In discrete space with basis $|x_i\rangle$,

- Orthogonality: $\langle x_i | x_j \rangle = \delta_{ij}$
- Completeness: $\sum_{i=1}^n |x_i\rangle \langle x_i| = I$

Natural generalization to continuous functions.

- Orthogonality: $\langle x | x' \rangle = \delta(x - x')$
- Completeness: $\int_a^b |x\rangle \langle x| dx = I$

δ is the Dirac delta function, defined as

$$\begin{aligned} \delta(x - x') &= 0, \quad x \neq x' \\ \int_a^b \delta(x - x') dx' &= 1, \quad a < x < b \end{aligned}$$

Sometimes we write $\delta(0) = \infty$, but this isn't well defined except in the context of an integral. The δ also works like a "sampling function":

$$\int \delta(x - x') f(x') dx' = f(x)$$

Derivative: The derivative of the delta function acts like an operator

$$\delta'(x - x') = \delta(x - x') \frac{d}{dx}$$

To see this, note that

$$\int \delta'(x - x') f(x') dx' = \frac{d}{dx} \int \delta(x - x') f(x') dx' = \frac{df(x)}{dx}$$

Fourier Series: We can also represent the delta function using a fourier series. Recall that

$$\text{Fourier transform: } f(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

$$\text{Inverse transform: } f(x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx'} f(k) dk$$

$$f(x') = \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x'-x)} dk \right) f(x) dx$$

Reasoning from the use of delta as a sampling function,

$$\delta(x' - x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x'-x)} dk$$

1.2 The Derivative Operator

We can have operators in infinite dimensions, e.g.

$$\Omega |f\rangle = |\tilde{f}\rangle$$

The derivative is one such operator:

$$D |f\rangle = \left| \frac{df}{dx} \right\rangle$$

The elements of D in the x basis are

$$D_{xx'} \equiv \langle x | D | x' \rangle = \delta'(x - x') = \delta(x - x') \frac{d}{dx'}$$

To see this, recall the use of δ as a sampling function.

$$\begin{aligned} \langle x | D | f \rangle &= \langle x | \frac{df}{dx} \rangle = \frac{df(x)}{dx} \\ \int \langle x | D | x' \rangle \langle x' | f \rangle dx' &= \frac{df}{dx} \end{aligned}$$

1.2.1 The Hermitian Operator $K = -iD$

However, D is not Hermitian: $D_{xx'} \neq D_{x'x}^*$.

$$\begin{aligned} D_{xx'} &= \delta'(x - x') \\ D_{x'x}^* &= \delta'(x' - x)^* = -\delta'(x - x') \end{aligned}$$

Instead, we can make D Hermitian using a constant complex term: $K = -iD$. Then

$$\begin{aligned} K_{xx'} &= -i\delta'(x - x') \\ K_{x'x}^* &= [-i\delta'(x' - x)]^* = i\delta'(x' - x) = -i\delta(x - x') \end{aligned}$$

Warning: For operators in infinite dimensions, $K_{xx'} = K_{x'x}^*$ doesn't actually guarantee Hermiticity. To see this, note that a Hermitian operator must satisfy (for functions f, g)

$$\begin{aligned} \langle g|K|f\rangle &= \langle f|K|g\rangle^* \\ \int_a^b \int_a^b \langle g|x\rangle \langle x|K|x'\rangle \langle x'|f\rangle dx dx' &\stackrel{?}{=} \left(\int_a^b \int_a^b \langle f|x\rangle \langle x|K|x'\rangle \langle x'|g\rangle dx dx' \right)^* \\ \int_a^b g^*(x) \left[-i \frac{df(x)}{dx} \right] dx &\stackrel{?}{=} \left(\int_a^b f^*(x) \left[-i \frac{dg(x)}{dx} \right] dx \right)^* \\ -ig^*(x)f(x) \Big|_a^b + i \int_a^b \frac{dg^*(x)}{dx} f(x) dx &\neq i \int_a^b \frac{dg^*(x)}{dx} f(x) dx \end{aligned}$$

In order to be Hermitian, the **surface term** of K must be 0. K is Hermitian in the space of functions where this holds. $K_{xx'} = K_{x'x}^*$ is no longer a sufficient condition.

$$-ig^*(x)f(x) \Big|_a^b = 0$$

Examples where

- String clamped at one end, $f(0) = f(L) = 0$. Derivative is $K = -i \frac{d}{dx}$
- Periodic function, $f(\phi) = f(\phi + 2\pi)$. Derivative is $K = -i \frac{d}{d\phi}$

In QM, we study functions $-\infty < x < \infty$. There are two classes of such functions.

1. f vanishes as $x \rightarrow -\infty$ and $x \rightarrow \infty$. Here, $K = -i \frac{d}{dx}$ is Hermitian.
2. f doesn't vanish, like e^{ikx} . If K is Hermitian, it must satisfy

$$e^{ikx} e^{ik'x} \Big|_{-\infty}^{\infty} = 0$$

If $k = k'$, this holds. Otherwise, we define the limit as $|x| \rightarrow \infty$ to be the average over a large interval. Then if $k \neq k'$,

$$\lim_{x \rightarrow \infty} e^{ikx} e^{ik'x} = \lim_{L, \Delta \rightarrow \infty} \frac{1}{\Delta} \int_L^{L+\Delta} e^{i(k-k')x} dx = 0$$

Thus K is also Hermitian for class 2 functions.

1.2.2 Eigenvalues of K

For eigenvalue k ,

$$\begin{aligned} K|k\rangle &= k|k\rangle \\ \langle x|K|k\rangle &= k\langle x|k\rangle \\ \int \langle x|K|x'\rangle \langle x'|k\rangle dx' &= k\langle x|k\rangle \end{aligned}$$

Let $\langle x|k\rangle = \psi_k(x)$. Using the definition $K = -iD$,

$$-i \frac{d}{dx} \psi_k(x) = k \psi_k(x)$$

The solution is $\psi_k(x) = A e^{ikx}$. Thus, any real number k is an eigenvalue of K with eigenvector $A e^{ikx}$. Since A is a free parameter, set it to $1/\sqrt{2\pi}$ so that

$$|k\rangle \leftrightarrow \frac{1}{\sqrt{2\pi}} e^{ikx}$$

$$\langle k|k'\rangle = \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|k'\rangle dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k'-k)x} dx = \delta(k' - k)$$

Thus $\langle k|k'\rangle = 0$ if $k \neq k'$, and blows up if $k = k'$.

We require a real eigenvalue k because if k is imaginary, e^{ikx} blows up as $x \rightarrow \infty$. The choice of a real eigenvalue restricts us to the **physical Hilbert space**.

1.2.3 Conjugate actions of X and K

In finite dimensions, the eigenfunctions of a Hermitian operator form a complete basis. Assume this holds for infinite dimensions.

K is Hermitian. Expand in both X and K basis. Recall $|k\rangle = \frac{1}{\sqrt{2\pi}} e^{ikx}$.

$$f(k) = \langle k|f\rangle = \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|f\rangle dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

$$f(x) = \langle x|f\rangle = \int_{-\infty}^{\infty} \langle x|k\rangle \langle k|f\rangle dk = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} f(k) dk$$

The Fourier transform pass f from one complete basis to another.

Additionally, note the matrix elements of X and K in the X and K basis, respectively:

$$\langle x|X|x'\rangle = x' \delta(x - x')$$

$$\langle k|K|k'\rangle = k' \delta(k - k')$$

In finite dimensions, we had (in the X basis) $X|x\rangle = x|x\rangle$. To see the action of X in infinite dimensions, let $X|f\rangle = |\tilde{f}\rangle$.

$$\langle x|X|f\rangle = \int_{-\infty}^{\infty} \langle x|X|x'\rangle \langle x'|f\rangle dx' = x f(x)$$

$$\langle x|X|f\rangle = \langle x|\tilde{f}\rangle = \tilde{f}(x)$$

Thus $\tilde{f}(x) = x f(x)$, and the effect of X in infinite dimensions is $X|f(x)\rangle = |x f(x)\rangle$. That is, multiplying $f(x)$ by x .

An interesting phenomenon occurs if we calculate the matrix elements of X in the K basis. Recall $|k\rangle = \frac{1}{\sqrt{2\pi}} e^{ikx}$.

$$\begin{aligned}\langle k|X|k'\rangle &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|X|x'\rangle \langle x'|k'\rangle dx dx' \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x' \delta(x-x') \langle k|x\rangle \langle x'|k'\rangle dx dx' \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} e^{ik'x} dx \\ &= i \frac{d}{dk} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k'-k)x} dx \right) = i \delta'(k-k')\end{aligned}$$

Thus, if g is a function with image $g(k)$ in the K basis,

$$X |g(k)\rangle = |i \frac{dg(k)}{dk}\rangle$$

To summarize,

$$\text{In } X \text{ basis } \begin{cases} X \equiv x \\ K \equiv -i \frac{d}{dx} \end{cases}$$

$$\text{In } K \text{ basis } \begin{cases} X \equiv i \frac{d}{dx} \\ K \equiv k \end{cases}$$

Operators like these are **conjugate** to each other.

Note that X and K do not commute.

$$\begin{aligned}XK|f\rangle &= -ix \frac{df(x)}{dx} \\ KX|f\rangle &= -i \frac{d}{dx} (xf(x)) = -ix \frac{df(x)}{dx} - if(x)\end{aligned}$$

Thus $[X, K]|f\rangle = if = iI|f\rangle$. Since f was arbitrary, we have $[X, K] = iI$.

In QM, X and K are the main operators we'll be seeing (X gives position, K gives derivative). However, they take different forms: $\hat{X} = X$ and $\hat{P} = \hbar K$.

2 Review of Classical Mechanics

2.1 Lagrangian Mechanics

Newtonian mechanics can find the path of a particle at an infinitesimal time slice:

$$m\ddot{x} = -\frac{dV}{dx}$$

Given $x(t_i)$ and $\dot{x}(t_i)$ at a point t_i , we can obtain the classical path's $x_{\text{cl}}(t_i + \Delta t)$ and $\dot{x}_{\text{cl}}(t_i + \Delta t)$ after some infinitesimal time Δt using the equation above. By continuing this process, we recreate the classical path x_{cl} of the particle.

$$\begin{aligned}x_{\text{cl}}(t_i + \Delta t) &= x(t_i) + \dot{x}(t_i)\Delta t \\ \dot{x}_{\text{cl}}(t_i + \Delta t) &= \dot{x}(t_i) + \ddot{x}(t_i)\Delta t\end{aligned}$$

Lagrangian mechanics seeks to find the classical path immediately. Given two points (x_i, t_i) and (x_f, t_f) , we can find the path taken by the particle between them. But what distinguishes the classical path from all other possible ones?

2.1.1 Procedure

1. Define the Lagrangian $L = T - V$. Assume time-independent potential, thus $L = L(x, \dot{x})$.
2. For each path from (x_i, t_i) to (x_f, t_f) , define the action as a function of the entire path:

$$S[x(t)] = \int_{t_i}^{t_f} L(x, \dot{x}) dt$$

3. **Principle of least action:** the classical path is that on which S is minimized.

NB: assumption of a time-independent potential doesn't always hold. For example, $F_B = qv \times B$. So $L = T - V$ isn't always the correct form.

2.1.2 The Euler-Lagrange Equation

For finite variables, the function was at a extremal if the 1st order change δf vanishes:

$$\begin{aligned}f(\mathbf{x}^0 + \eta) &= f(\mathbf{x}^0) + \sum_{i=1}^n \left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}^0} \eta_i + O(\eta^2) \\ \delta f &= \sum_{i=1}^n \left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}^0} \eta_i = 0\end{aligned}$$

Assume this holds for functions of infinite variables, like $S[x(t)]$. Let $x_{\text{cl}}(t)$ be the classical path, and $x_{\text{cl}}(t) + \eta(t)$ a nearby path. Endpoints must match, so $\eta(t_i) = \eta(t_f) = 0$. Now find the extremal where 1st order change δS vanishes:

$$\begin{aligned}S[x_{\text{cl}}(t) + \eta(t)] &= \int_{t_i}^{t_f} L(x_{\text{cl}}(t) + \eta(t), \dot{x}_{\text{cl}}(t) + \dot{\eta}(t)) dt \\ &= \int_{t_i}^{t_f} \left[L(x_{\text{cl}}(t), \dot{x}_{\text{cl}}(t)) + \left. \frac{\partial L}{\partial x(t)} \right|_{x_{\text{cl}}} \cdot \eta(t) + \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{\text{cl}}} \cdot \dot{\eta}(t) + \dots \right] dt \\ &\approx S[x_{\text{cl}}(t)] + \delta S\end{aligned}$$

Find where $\delta S = 0$ for an arbitrary deviation η :

$$\begin{aligned}
 0 = \delta S &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} \right|_{x_{cl}} \cdot \eta(t) + \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \dot{\eta}(t) \right] dt \\
 &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} \right|_{x_{cl}} \cdot \eta(t) \right] dt + \left[\left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \eta(t) \right]_{t_i}^{t_f} - \int_{t_i}^{t_f} \left[\frac{d}{dt} \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \eta(t) \right] dt \\
 &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \right] \cdot \eta(t) dt
 \end{aligned}$$

Thus we get the Euler-Lagrange equation, which must be satisfied by a classical path:

$$\left[\frac{\partial L}{\partial x(t)} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}(t)} \right]_{x_{cl}} = 0$$

2.1.3 Conservation Laws from the Lagrangian

A benefit of the Lagrangian is that unlike the Newtonian, coordinates don't have to be Cartesian. We can have an arbitrary q_i and $\dot{q}_i$.

Further note:

- Canonical momentum conjugate: $p_i = \partial L / \partial \dot{q}_i$
- Canonical force conjugate: $F_i = \partial L / \partial q_i$

$$F_i = \frac{d}{dt} p_i$$

E.g. if $q_i = \theta$, then p_i is angular momentum and F_i is torque.

Call q_i a **cyclic coordinate** if L depends on $\dot{q}_i$, but not q_i . Then p_i is conserved:

$$\frac{dp_i}{dt} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} = 0$$

Conservation laws fall out naturally out of the Lagrangian, while it may be opaque in the Newtonian. For example, consider a Lagrangian in polar coordinates $L = L(\rho, \phi)$. The metric is $ds^2 = d\rho^2 + \rho d\phi^2$. Thus the infinitesimal velocity is

$$v = \frac{ds}{dt} = \sqrt{(\dot{\rho})^2 + (\rho \dot{\phi})^2}$$

The Lagrangian is

$$L = \frac{1}{2} m [\dot{\rho}^2 + \rho^2 \dot{\phi}^2] - V(\rho, \phi)$$

Applying Euler-Lagrange for ρ and ϕ respectively gives

$$\begin{aligned}
 \frac{d}{dt} (m\dot{\rho}) &= -\frac{\partial V}{\partial \rho} + m\rho\dot{\phi}^2 \quad \longrightarrow \quad m\ddot{\rho} = -\frac{\partial V}{\partial \rho} + m\rho(\dot{\phi})^2 \\
 \frac{d}{dt} (m\rho^2\dot{\phi}) &= -\frac{\partial V}{\partial \phi} \quad \longrightarrow \quad m\ddot{\phi} = -\frac{1}{\rho^2} \frac{\partial V}{\partial \phi} - \frac{1}{\rho} (2m\dot{\rho}\dot{\phi})
 \end{aligned}$$

Note how easily we can derive the centrifugal and coriolis forces. Also note that conservation laws for cyclic coordinates $V(\rho, \phi) = V(\rho)$ and $V(\rho, \phi) = V(\phi)$ fall out naturally, and the corresponding equations of motion are that of $\partial V/\partial \phi = 0$ and $\partial V/\partial \rho = 0$.

In sum:

- Equations of motion are simpler to find with Lagrangian
- Conservation laws are easy to see from Lagrangian
- Path integral formalism uses Lagrangian (later)

2.1.4 The Electromagnetic Lagrangian

Electromagnetic force:

$$\vec{F} = q \left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B} \right)$$

Note: this is Gaussian units, hence the $1/c$ factor for magnetic force.

The electromagnetic force is velocity-dependent, so $L = T - V$ doesn't work. Instead, we construct L to reduce to the expected equations. The EM lagrangian is

$$L_{\text{EM}} = \frac{1}{2} m \vec{v} \cdot \vec{v} - q\phi + \frac{q}{c} \vec{v} \cdot \vec{A}$$

ϕ and $\vec{A}$ are the scalar and vector potentials, related to E and B by

$$\begin{aligned} \vec{E} &= -\nabla\phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} \\ \vec{B} &= \nabla \times \vec{A} \end{aligned}$$

The Euler-Lagrange equation is

$$\begin{aligned} \frac{d}{dt} \left(m\dot{x}_i + \frac{q}{c} A_i \right) &= -q \frac{\partial \phi}{\partial x_i} + \frac{q}{c} \frac{\partial (\vec{v} \cdot \vec{A})}{\partial x_i}, \quad i = 1, 2, 3 \\ \frac{d}{dt} \left(m\vec{v} + \frac{q}{c} \vec{A} \right) &= -q \nabla \phi + \frac{q}{c} \nabla (\vec{v} \cdot \vec{A}) \\ \frac{d}{dt} (m\vec{v}) &= -q \nabla \phi + \frac{q}{c} \left[-\frac{d\vec{A}}{dt} + \nabla (\vec{v} \cdot \vec{A}) \right] \end{aligned} \tag{1}$$

By time-dependence of the magnetic vector potential $\vec{A}$,

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + \sum_i \frac{\partial \vec{A}}{\partial x_i} \frac{dx_i}{dt} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \nabla) \vec{A}$$

Thus equation (1) becomes

$$\frac{d}{dt} (m\vec{v}) = -q \nabla \phi - \frac{q}{c} \frac{\partial \vec{A}}{\partial t} + \frac{q}{c} \underbrace{\left[\nabla (\vec{v} \cdot \vec{A}) - (\vec{v} \cdot \nabla) \vec{A} \right]}_{\vec{v} \times (\nabla \times \vec{A})} \tag{2}$$

This reduces to $\vec{F} = q\vec{E} + \frac{q}{c}\vec{v} \times \vec{B}$.

The time-dependent EM field is usually not conservative, so there's no path-independent work function to use as the potential. Thus L_{EM} is made to fit the situation and reduce as expected. $L = T - V$ only works for conservative forces.

2.1.5 The Two-Body Problem

Assume 2 masses m_1, m_2 that exert equal and opposite forces on each other. We can reduce this to a one-body problem with Lagrangian mechanics.

The potential only depends on relative position, $V(\vec{r}_1, \vec{r}_2) = V(\vec{r}_1 - \vec{r}_2)$. Thus $6 \rightarrow 3$ cyclic coordinates in 3D.

The two-body Lagrangian is

$$L = \frac{1}{2}m_1|\dot{\vec{r}}_1|^2 + \frac{1}{2}m_2|\dot{\vec{r}}_2|^2 - V(r_1 - r_2) \quad (1)$$

Define $\vec{r}_{\text{CM}}$ and $\vec{r}$ such that

$$\begin{aligned} \vec{r}_{\text{CM}} &= \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}, \quad \vec{r} = \vec{r}_1 - \vec{r}_2 \\ \vec{r}_1 &= \vec{r}_{\text{CM}} + \frac{m_2\vec{r}}{m_1 + m_2} \\ \vec{r}_2 &= \vec{r}_{\text{CM}} - \frac{m_1\vec{r}}{m_1 + m_2} \end{aligned}$$

The two-body Lagrangian becomes

$$L = \frac{1}{2}(m_1 + m_2)|\dot{\vec{r}}_{\text{CM}}|^2 + \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}|\dot{\vec{r}}|^2 - V(\vec{r}) \quad (2)$$

This can be thought of as the Lagrangian of two fictitious particles that don't interact: one "CM" particle with mass $M = m_1 + m_2$, and one "reduced mass" particle with mass $\mu = (m_1m_2)/(m_1 + m_2)$. Note that $L(r, \dot{r}, r_{\text{CM}}, \dot{r}_{\text{CM}}) = L(r, \dot{r}) + L(r_{\text{CM}}, \dot{r}_{\text{CM}})$.

By going to the CM frame, we can eliminate $\dot{r}_{\text{CM}}$ and just solve for the motion of μ . We've reduced the two-body problem to a one-body problem.

2.2 Hamiltonian Mechanics

2.2.1 Derivation

The Lagrangian uses $L(q, \dot{q})$. The Hamiltonian reverses the role of p and $\dot{q}$, using $H(q, p)$. This "role reversal" is achieved with a **Legendre transform**. Suppose $u(x) = df/dx$ and $x(u) = dg/du$. The relationship between f and g is a Legendre transform:

$$g(u) = x(u)u - f(x(u))$$

Define the Hamiltonian $H(q, p)$ as a Legendre transform of the Lagrangian $L(q, \dot{q})$:

$$H(q, p) = \sum_{i=1}^n p_i \dot{q}_i - L(q, \dot{q}) \quad , \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Hamilton's **canonical equations** can be found:

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad , \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Lagrangian	Hamiltonian
n coordinates q_i and n velocities $\dot{q}_i$	n coordinates q_i and n momenta p_i
State of system is point in n -dimensional configuration space	State of system is point in $2n$ -dimensional configuration space
n coordinates evolve according to n 2nd-order differential equations	$2n$ coordinates and momenta evolve according to $2n$ 1st-order differential equations
For given L , multiple trajectories can pass through a point in configuration space depending on $\dot{q}_i$.	For given H , only one trajectory passes through a point in configuration space.

For conservative forces with $T = \frac{1}{2}m\dot{x}_i^2$, we get

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m_i \dot{x}_i$$

$$H = \sum_i p_i \dot{x}_i - L(x, \dot{x}) = T + V$$

But this isn't always the right form. It's only for conservative forces.

2.2.2 The Electromagnetic Hamiltonian

If L_{EM} generates the correct equations of motion, then H_{EM} will as well. But it won't take the form $H = T + V$ since now the force isn't conservative.

Start from the electromagnetic Lagrangian:

$$L_{\text{EM}} = \frac{1}{2}m\vec{v} \cdot \vec{v} - q\phi + \frac{q}{c}\vec{v} \cdot \vec{A}$$

Now find the canonical momentum $\vec{p}$ and the Hamiltonian H :

$$\vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} + \frac{q}{c}\vec{A}$$

$$H_{\text{EM}} = \vec{p} \cdot \vec{v} - L_{\text{EM}} = \frac{1}{2}m\vec{v} \cdot \vec{v} + q\phi = \frac{|\vec{p} - q\vec{A}/c|^2}{2m} + q\phi$$

2.2.3 Conservation Laws from the Hamiltonian

If q_i is a cyclic coordinate in the Hamiltonian, then

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} = 0$$

The associated momentum is conserved. But other quantities are too, and this is often easier to see in the Hamiltonian than in the Lagrangian.

Let $\omega(p, q)$ be a state function with **no explicit dependence on t** . Define the **Poisson bracket**

$$\{\omega, \lambda\} = \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \right)$$

The time derivative of ω is:

$$\begin{aligned} \frac{d\omega}{dt} &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial q_i}{\partial t} + \frac{\partial \omega}{\partial p_i} \frac{\partial p_i}{\partial t} \right) \\ &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \\ &= \{\omega, H\} \end{aligned}$$

Any state function ω whose Poisson bracket with H vanishes is conserved. H itself is conserved if there's no explicit t dependence.

The Poisson bracket between q and p satisfy

$$\begin{aligned} \{q_i, q_j\} &= \{p_i, p_j\} = 0 \\ \{q_i, p_j\} &= \delta_{ij} \end{aligned}$$

From these, the canonical equations can be derived in terms of Poisson brackets:

$$\begin{aligned} \dot{q}_i &= \{q_i, H\} \\ \dot{p}_i &= \{p_i, H\} \end{aligned}$$

2.2.4 Canonical Transformations

Euler-Lagrange equations are invariant under a coordinate change $(q, \dot{q}) \rightarrow (\bar{q}, \dot{\bar{q}})$, which implies the invariance of Hamilton's equations under $(q, p) \rightarrow (\bar{q}, \bar{p})$.

General transformations of the form $q \rightarrow \bar{q}(q, p)$ and $p \rightarrow \bar{p}(q, p)$ are called **canonical transformations** if they obey the canonical equations. $(\bar{q}, \bar{p})$ are called **canonical coordinates**.

Theorem: If $(\bar{q}(q, p), \bar{p}(q, p))$ are canonical coordinates iff.

$$\begin{aligned} \{\bar{q}_j, \bar{q}_k\} &= 0 = \{\bar{p}_j, \bar{p}_k\} \\ \{\bar{q}_j, \bar{p}_k\} &= \delta_{jk} \end{aligned}$$

Proof: For state functions $\omega(q, p)$,

$$\dot{\omega} = \{\omega, H\} = \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial H}{\partial q_i} \right)$$

Thus for $\dot{q}_j$,

$$\dot{q}_j = \sum_i \left(\frac{\partial \bar{q}_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \bar{q}_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \quad (1)$$

Using the chain rule, we can also get

$$\begin{cases} \frac{\partial H(q, p)}{\partial p_i} = \frac{\partial H(\bar{q}, \bar{p})}{\partial p_i} = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \frac{\partial \bar{q}_k}{\partial p_i} + \frac{\partial H}{\partial \bar{p}_k} \frac{\partial \bar{p}_k}{\partial p_i} \right) \\ \frac{\partial H(q, p)}{\partial q_i} = \frac{\partial H(\bar{q}, \bar{p})}{\partial q_i} = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \frac{\partial \bar{q}_k}{\partial q_i} + \frac{\partial H}{\partial \bar{p}_k} \frac{\partial \bar{p}_k}{\partial q_i} \right) \end{cases} \quad (2)$$

Plugging (2) into (1) and simplifying gives (following a similar procedure for $\dot{p}_j$),

$$\begin{cases} \dot{q}_j = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \{\bar{q}_j, \bar{q}_k\} + \frac{\partial H}{\partial \bar{p}_k} \{\bar{q}_j, \bar{p}_k\} \right) \\ \dot{p}_j = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \{\bar{p}_j, \bar{q}_k\} + \frac{\partial H}{\partial \bar{p}_k} \{\bar{p}_j, \bar{p}_k\} \right) \end{cases} \quad (3)$$

For (3) to reduce to the canonical equations, we must have $\{\bar{q}_j, \bar{q}_k\} = 0 = \{\bar{p}_j, \bar{p}_k\}$ and $\{\bar{q}_j, \bar{p}_k\} = \delta_{jk}$.

*Note that the Poisson bracket is independent of parametrization. Whether $\omega(q, p)$ or $\omega(\bar{q}, \bar{p})$,

$$\{\omega, \lambda\}_{q, p} = \{\omega, \lambda\}_{\bar{q}, \bar{p}}$$

2.2.5 Active Transformations

The transformation $q \rightarrow \bar{q}(q, p)$ and $p \rightarrow \bar{p}(q, p)$ is passive if (q, p) and $(\bar{q}, \bar{p})$ refer to the same point in phase space.

If the transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$ preserves the range of variables, it's called **regular**.

- E.g. transformation from a Cartesian plane to a translated/rotated one is regular.
- But a transformation from a Cartesian plane to spherical coordinates isn't. Some variables are now bounded by 2π instead of ∞ .

A transformation that "shifts" the phase space so that (q, p) and $(\bar{q}, \bar{p})$ no longer refer to the same point is called **active**. Active transformations are a subset of regular ones.

We call ω invariant under the regular transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$ if

$$\omega(q, p) = \omega(\bar{q}, \bar{p})$$

2.2.6 Symmetries

Two results:

1. If H is invariant under the following infinitesimal transformations, then $g(q, p)$ is conserved. We call g a **generator of the transformation**.

$$\begin{aligned} q_i &\rightarrow \bar{q}_i + \varepsilon \frac{\partial g}{\partial p_i} \equiv q_i + \delta q_i \\ p_i &\rightarrow \bar{p}_i - \varepsilon \frac{\partial g}{\partial q_i} \equiv p_i + \delta p_i \end{aligned}$$

2. If H is invariant under the regular, canonical, but not necessarily infinitesimal transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$, and if (q, p) is a solution to the equations of motion, so is $(\bar{q}, \bar{p})$.

Proof of (1): Note that

$$\begin{aligned} \delta H &= \sum_i \frac{\partial H}{\partial q_i} (\delta q_i) + \frac{\partial H}{\partial p_i} (\delta p_i) \\ &= \sum_i \frac{\partial H}{\partial q_i} \left(\varepsilon \frac{\partial g}{\partial p_i} \right) + \frac{\partial H}{\partial p_i} \left(-\varepsilon \frac{\partial g}{\partial q_i} \right) \\ &= \varepsilon \{H, g\} \end{aligned}$$

For H to be invariant (to first order), $\delta H = 0$ and $\{H, g\} = 0$. Thus g is conserved.

In general for state variables ω ,

$$\delta \omega = \varepsilon \{\omega, g\}$$

2.2.7 A Relation Between S and E

Let x_{cl} be a classical path from (x_i, t_i) to (x_f, t_f) , $H(t_f)$ be the Hamiltonian at t_f , and S the classical action. A neat relation is

$$\frac{\partial S_{\text{cl}}(x_f, t_f; x_i, t_i)}{\partial t_f} = -H(t_f)$$

If E is conserved, this becomes

$$\frac{\partial S_{\text{cl}}(x_f, t_f; x_i, t_i)}{\partial t_f} = -E$$

Proof: Hold x_i, x_f fixed and vary with $\eta(t)$. For convenience, $x_i = t_i = 0$ and $\eta(0) = 0$.

$$x(t) = x_{\text{cl}}(t) + \eta(t)$$

The variation in action can be split into (1) the contribution from difference in path, (2) the contribution from the new “end” time $t = t_f + \Delta t$.

$$\begin{aligned}
 \delta S_{\text{cl}} &= \int_0^{t_f} \frac{\partial L}{\partial t} dt + L(t_f) \Delta t \\
 &= \int_0^{t_f} \left[\frac{\partial L}{\partial x} \eta(t) + \frac{\partial L}{\partial \dot{x}} \dot{\eta}(t) \right] dt + L(t_f) \Delta t \\
 &= \int_0^{t_f} \left[-\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} + \frac{\partial L}{\partial x} \right] \eta(t) dt + \int_0^{t_f} \frac{d}{dt} \left[\frac{\partial L}{\partial \dot{x}} \eta(t) \right] dt + L(t_f) \Delta t \\
 &= 0 + \left[\frac{\partial L}{\partial \dot{x}} \eta(t) \right]_0^{t_f} + L(t_f) \Delta t
 \end{aligned}$$

Substitute $\eta(t_f) = -\dot{x}(t_f) \Delta t$ to get

$$\delta S = \left[-\frac{\partial L}{\partial \dot{x}} \dot{x} + L \right]_{t_f} \Delta t = -H(t_f) \Delta t$$

2.3 A Problem: Wave-Particle Duality

2.3.1 Plane Waves

An electromagnetic wave ψ that travels at c obey the wave equation

$$\nabla^2 \psi = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

A special case is the plane wave:

$$\psi(x, t) = A e^{i\phi} = A \exp \left(i \left(\frac{2\pi}{\lambda} x - \frac{2\pi}{T} t \right) \right)$$

- Wave number $k = 2\pi/\lambda$ (phase change per unit length at fixed t)
- Angular frequency $\omega = 2\pi/T$ (phase change per unit time at fixed x)
- Wave speed $v = \omega/T$

This can be extended to plane waves in 3D, using the **wave vector** $\vec{k}$.

$$\psi(\vec{r}, t) = A e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad , \quad \omega = |\vec{k}|v$$

2.3.2 Double-Slit Experiment

The wave prediction is $I_{1+2} \neq I_1 + I_2$, due to interference effects.

The classical particle prediction is $I_{1+2} = I_1 + I_2$.

By opening one slit and lowering the intensity, we find that light energy comes in bundles. Each bundle has energy and momentum that satisfies

$$E = \hbar\omega \quad , \quad p = \hbar k$$

Thus light seems to be made up of particles called photons. But if both slits are opened at I , we get a wave effect: $I_{1+2} \neq I_1 + I_2$. The difference can't be reconciled classically.

Born suggested that each photon is associated with a wave ψ called the **probability amplitude**. $|\psi(x)|^2$ is the probability of finding the particle at x . The wave is described by

$$\omega = \frac{E}{\hbar} \quad , \quad k = \frac{p}{\hbar}$$

The wave interferes with itself (after being split by the double-slit) to form the overall pattern according to $|\psi(x)|^2$.

De Broglie suggested that every particle of momentum p has a wave associated with it, not just massless photons. The wave number is

$$k = \frac{p}{\hbar}$$

3 Postulates of Quantum Mechanics

3.1 The Postulates

Classical:

1. State of particle is described by $x(t)$ and $p(t)$ in 2D phase space.
2. Every dynamical variable $\omega(x, p)$ is a function of x, p .
3. If a particle is in the state $(x; p)$, a perfect measurement of ω yields $\omega(x, p)$ without changing the state.
4. Time evolution described by

$$\dot{x} = \frac{\partial H}{\partial p} \quad , \quad \dot{p} = -\frac{\partial H}{\partial x}$$

Quantum:

1. State of particle is a vector $|\psi(t)\rangle$ in Hilbert space.
2. Independent variables x, p of classical mechanics become Hermitian operators X, P such that

$$\begin{aligned} \langle x|X|x'\rangle &= x\delta(x-x') \\ \langle x|P|x'\rangle &= -\hbar\delta'(x-x') \end{aligned}$$

(Note that this is just X and $P = \hbar K$ from before). Then $\omega(x, p)$ becomes

$$\Omega(X, P) = \omega(x \rightarrow X, p \rightarrow P)$$

3. If a particle is in the state $|\psi\rangle$, a perfect quantum measurement of Ω yields an eigenvalue ω with probability

$$P(\omega) \propto |\langle \omega|\psi\rangle|^2$$

The state of the system changes:

$$|\psi\rangle \rightarrow |\omega\rangle$$

Note that Ω is Hermitian, so the eigenvalues are real. In terms of the projection operator $\mathbb{P}_\omega = |\omega\rangle\langle\omega|$,

$$\begin{aligned} P(\omega) &\propto |\langle\omega|\psi\rangle|^2 = \langle\psi|\omega\rangle\langle\omega|\psi\rangle \\ &= \langle\psi|\mathbb{P}_\omega|\psi\rangle \\ &= \langle\psi|\mathbb{P}_\omega\mathbb{P}_\omega|\psi\rangle \\ &= \langle\mathbb{P}_\omega\psi|\mathbb{P}_\omega\psi\rangle \end{aligned}$$

4. Time evolution described by

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

$H(X, P) = H(x \rightarrow X, p \rightarrow P)$ from the classical Hamiltonian.

Potential problems:

1. $\Omega(x \rightarrow X, p \rightarrow P)$ is ambiguous. E.g. for $\omega = xp$, is $\Omega = XP$ or PX ? Solution is to symmetrize with $\Omega = (XP + PX)/2$.
2. Ω can be degenerate, e.g. two eigenvalues $\omega_1 = \omega_2 = \omega$. Then what is $P(\omega)$? Solution: find some basis $|\omega_1\rangle, |\omega_2\rangle$ in the eigenspace $\mathbb{V}_\omega$. Then

$$P(\omega) = |\langle\omega_1|\psi\rangle|^2 + |\langle\omega_2|\psi\rangle|^2$$

This is the same result we get by assuming ω_1, ω_2 are infinitesimally distinct, then asking for $P(\omega_1 \text{ or } \omega_2)$.

3. Eigenvalue spectrum of Ω can be continuous, e.g. X . Solution: expand

$$|\psi\rangle = \int |\omega\rangle \langle\omega|\psi\rangle d\omega$$

Introduce the **wave function in ω space** $\psi(\omega) = \langle\omega|\psi\rangle$ and the **probability density** $P(\omega) = |\langle\omega|\psi\rangle|^2$. $P(\omega) d\omega$ is the probability of measuring between ω and $\omega + d\omega$. Note $\int P(\omega) d\omega = 1$.

3.1.1 Ensembles

Used to test quantum theory. A quantum ensemble is a large number N of particles all in state $|\omega\rangle$.

A single particle, after measurement, could give any of the eigenvalues. Measure all N to get a statistical result.

Difference with classical:

- In classical ensemble of N particles, if we measure λ_1 in $N/3$ trials and λ_2 in $2N/3$ trials, we expect there were $N/3$ amount of λ_1 and $2N/3$ amount of λ_2 before measurement.
 - In quantum ensemble, all particles are assumed to be $|\omega\rangle$ before measurement.
-

To calculate average over ensemble, use the **expectation value**:

$$\begin{aligned}
\langle \Omega \rangle &= \sum_i P(\omega_i) \omega_i = \sum_i |\langle \omega_i | \psi \rangle|^2 \omega_i \\
&= \sum_i \langle \psi | \omega_i \rangle \langle \omega_i | \psi \rangle \omega_i \\
&= \sum_i \langle \psi | \Omega | \omega_i \rangle \langle \omega_i | \psi \rangle \\
&= \langle \psi | \Omega | \psi \rangle
\end{aligned}$$

To measure the uncertainty (fluctuation around the mean), use the **standard deviation**:

$$\Delta \Omega = \langle (\Omega - \langle \Omega \rangle)^2 \rangle^{1/2}$$

Classically, this is called “root-mean-square” deviation. In QM, it’s “uncertainty in Ω .”

$$(\Delta \Omega)^2 = \begin{cases} \sum_i P(\omega_i) (\omega_i - \langle \Omega \rangle)^2 & \text{(discrete spectrum)} \\ \int P(\omega) (\omega - \langle \Omega \rangle)^2 d\omega & \text{(continuous spectrum)} \end{cases}$$

E.g. if a particle has $\langle X \rangle = a$ and $\Delta X = \Delta$, then we know it’s likely to be spotted near a with deviation of order Δ .

3.1.2 Compatible Variables

Is there a filtering process by which we take $|\psi\rangle$ and produce a state with well-defined ω , λ for 2 variables Ω , Λ ?

- Not generally, because measurement changes the state. I.e. (if we do Ω then Λ)

$$\begin{aligned}
\text{first: } |\psi\rangle &\rightarrow \sum |\omega\rangle \langle \omega | \psi \rangle \\
\text{second: } |\omega\rangle &\rightarrow \sum |\lambda\rangle \langle \lambda | \omega \rangle
\end{aligned}$$

But if Ω and Λ have simultaneous eigenkets, then

$$\begin{aligned}
\Omega |\omega\lambda\rangle &= \omega |\omega\lambda\rangle \\
\Lambda |\omega\lambda\rangle &= \lambda |\omega\lambda\rangle \\
(\Omega\Lambda - \Lambda\Omega) |\omega\lambda\rangle &= 0
\end{aligned}$$

We can “know” both variables at the same time, if it has a simultaneous eigenket. Variables such that $[\Omega, \Lambda] = 0$ are called **compatible**.

E.g. The canonical commutation relation $[X, P] = i\hbar \neq 0$. Thus, a state can’t have well-defined eigenkets of X and P simultaneously. This is the Heisenberg uncertainty relation.

3.1.3 Density Matrices

Instead of an ensemble of N particles all in $|\psi\rangle$, we can consider an ensemble of N systems, n_i of which are $|i\rangle$.

Let $p_i = n_i/N$ be the probability of measuring a $|i\rangle$. Define the density matrix:

$$\rho = \sum_i p_i |i\rangle \langle i|$$

- “Pure”: all $n_i = 0$ except for one
- “Mixed”: otherwise.

The ensemble average of Ω is $\langle \bar{\Omega} \rangle = \sum_i p_i \langle i | \Omega | i \rangle$. Note that

$$\begin{aligned} \text{Tr}(\Omega \rho) &= \sum_j \langle j | \Omega \rho | j \rangle = \sum_i \sum_j \langle j | \Omega \rho | i \rangle \langle i | j \rangle p_i \\ &= \sum_i \sum_j \langle i | j \rangle \langle j | \Omega \rho | i \rangle p_i = \sum_i \langle i | \Omega \rho | i \rangle p_i \\ &= \langle \bar{\Omega} \rangle \end{aligned}$$

E.g. if we want the probability of obtaining ω , $\overline{P(\omega)}$, then

$$\begin{aligned} P(\omega) &= |\langle \omega | \psi \rangle|^2 = \langle \psi | \omega \rangle \langle \omega | \psi \rangle = \langle \psi | \mathbb{P}_\omega | \psi \rangle \\ \overline{P(\omega)} &= \text{Tr}(\mathbb{P}_\omega \rho) \end{aligned}$$

Features:

- $\rho^\dagger = \rho$
- $\rho^2 = \rho$ for pure state
- $\rho = \frac{1}{k} I$ for mixed state uniformly distributed over k sub-states
- $\text{Tr}(\rho) = 1$
- $\text{Tr}(\rho^2) \leq 1$, equality if ρ is pure.

3.2 Example: Gaussian Wavefunction

We'll find the Heisenberg uncertainty relation (sort of) for this model.

Put the wavefunction in the X -basis:

$$|\psi\rangle = \int_{-\infty}^{\infty} |x\rangle \langle x | \psi \rangle dx = \int_{-\infty}^{\infty} |x\rangle \psi(x) dx$$

Assume $\psi(x)$ is Gaussian and normalize to unity:

$$\psi(x) = \frac{1}{(\pi \Delta^2)^{1/4}} e^{-(x-a)^2/(2\Delta^2)}$$

In the X -basis, the effect of X is to multiply by x . I.e.

$$\begin{aligned}\langle x|X|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|X|x'\rangle \langle x'|\psi\rangle dx' = \int_{-\infty}^{\infty} x\delta(x-x')\psi(x') dx' \\ &= x\psi(x)\end{aligned}$$

Expectation value is $\langle X \rangle = a$ (which makes sense; $\psi(x)$ is centered around a).

Also, $\langle X^2 \rangle = \Delta^2/2 + a^2$

Then the uncertainty is

$$\begin{aligned}\Delta X &= \left[\langle \psi | (X - \langle X \rangle)^2 | \psi \rangle \right]^{1/2} = \left[\langle X^2 \rangle - 2a\langle X \rangle + a^2 \right]^{1/2} \\ &= \left[\langle X^2 \rangle - a^2 \right]^{1/2} \\ &= \frac{\Delta}{\sqrt{2}}\end{aligned}$$

Now for P . The action in the X basis is $-i\hbar \frac{d}{dx}$. I.e.

$$\begin{aligned}\langle x|P|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|P|x'\rangle \langle x'|\psi\rangle dx' = \int_{-\infty}^{\infty} -i\hbar\delta'(x-x')\psi(x') dx' \\ &= -i\hbar \frac{d\psi}{dx}\end{aligned}$$

Now find the eigenvalues. Note that $\langle x|p\rangle = \psi_p(x) \neq \psi(x)$.

$$\begin{aligned}P|p\rangle &= p|p\rangle \longrightarrow \langle x|P|p\rangle = p\langle x|p\rangle \\ -i\hbar \frac{d\psi_p(x)}{dx} &= p\psi_p(x) \\ \psi_p(x) &= \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}\end{aligned}$$

By finding $\psi(p) = \langle p|\psi\rangle$, we get a Gaussian of width $\hbar/(\Delta\sqrt{2})$ centered at 0. Thus $\langle P \rangle = 0$ and $\Delta P = \hbar/(\Delta\sqrt{2})$.

Put together: for this Gaussian wavefunction model, we get

$$\Delta X \cdot \Delta P = \frac{\hbar}{2}$$

This is a lower bound on the Heisenberg uncertainty relation.

3.3 Generalization to N Degrees of Freedom

Use mutually commuting operators $X_1, \dots, X_N$. In their simultaneous eigenbasis $|x_1, x_2, \dots, x_N\rangle$,

$$\langle x_1, x_2, \dots, x_N | x'_1, x'_2, \dots, x'_N \rangle = \delta(x_1 - x'_1) \cdots \delta(x_N - x'_N)$$

And

$$\begin{aligned} |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | \psi \rangle = \psi(x_1, \dots, x_N) \\ X_i |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | X_i | \psi \rangle = x_i \psi(x_1, \dots, x_N) \\ P_i |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | P_i | \psi \rangle = -i\hbar \frac{\partial}{\partial x_i} \psi(x_1, \dots, x_N) \end{aligned}$$

Then follow the procedure of replacing classical variables with quantum operators. E.g. for the classical operator

$$\omega = \frac{p_x^2 + p_y^2 + p_z^2}{2m} + x^2 + y^2 + z^2$$

We associate the quantum operator such that (with $X_1 = X$, $X_2 = Y$, and $X_3 = Z$)

$$\Omega |\omega\rangle = \omega |\omega\rangle \quad \Omega = \frac{P_x^2 + P_y^2 + P_z^2}{2m} + X^2 + Y^2 + Z^2$$

Then substitue $|\omega\rangle = \psi_\omega(x, y, z)$, $X \rightarrow x$, and $P_x \rightarrow -i\hbar \frac{d}{dx}$.

$$\left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + x^2 + y^2 + z^2 \right] \psi_\omega(x, y, z) = \omega \psi_\omega(x, y, z) \quad (1)$$

Once we have this form, we can change to other coordinates, e.g. spherical. Use

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \equiv \nabla^2 \equiv \frac{1}{r^2} \left(\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)$$

And substitute into (1) to get

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi_\omega}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi_\omega}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi_\omega}{\partial \phi^2} \right] + r^2 \psi_\omega = \omega \psi_\omega \quad (2)$$

3.4 The Schrödinger Equation

3.4.1 Hamiltonian Setup

Simply substitute $x \rightarrow X$, $p \rightarrow P$ in the classical Hamiltonian. E.g. for (1) harmonic oscillator, (2) constant force f , and (3) particle of charge q in an EM field,

$$\begin{aligned} 1) \quad H &= \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 &\longrightarrow H &= \frac{P^2}{2m} + \frac{1}{2} m \omega^2 X^2 \\ 2) \quad H &= \frac{p^2}{2m} - f x &\longrightarrow H &= \frac{P^2}{2m} - f X \\ 3) \quad H &= \frac{|\vec{p} - \frac{q}{c} \vec{A}(\vec{r}, t)|^2}{2m} + q \phi(\vec{r}, t) &\longrightarrow H &= \frac{1}{2m} \left(P \cdot P - \frac{q}{c} A \cdot P - \frac{q}{c} P \cdot A + \frac{q^2}{c^2} A \cdot A \right) + q \phi \end{aligned}$$

Note: P doesn't commute with A (which depends on X, Y, Z).

3.4.2 Time-Independent Schrödinger Equation

Start from Postulate IV, $i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle$. We seek to find a propagator of the form

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

Use the energy basis, since $H |E\rangle = E |E\rangle$.

$$|\psi(t)\rangle = \sum_E |E\rangle \langle E|\psi(t)\rangle \equiv \sum_E a_E(t) |E\rangle$$

Apply $(i\hbar \frac{\partial}{\partial t} - H)$ and use $(i\hbar \frac{\partial}{\partial t} - H) |\psi(t)\rangle = 0$ to get

$$\begin{aligned} 0 &= \sum_E (i\hbar \frac{\partial}{\partial t} - H) a_E(t) |E\rangle = \sum_E (i\hbar \dot{a}_E - E a_E) |E\rangle \\ a_E(t) &= a_E(0) e^{-iEt/\hbar} \\ \langle E|\psi(t)\rangle &= \langle E|\psi(0)\rangle e^{-iEt/\hbar} \end{aligned}$$

Now plug back in to $|\psi\rangle$ to get the propagator:

$$\begin{aligned} |\psi(t)\rangle &= \sum_E e^{-iEt/\hbar} |E\rangle \langle E|\psi(0)\rangle \\ U(t) &= \sum_E e^{-iEt/\hbar} |E\rangle \langle E| = e^{-iHt/\hbar} \end{aligned} \tag{1}$$

Because H is unitary, $U(t)$ is unitary, thus time evolution is unitary and the norm is preserved: $\langle \psi(t)|\psi(t)\rangle = \langle \psi(0)|\psi(0)\rangle$.

3.4.3 Time-Dependent Schrödinger Equation

To approximate a time-dependent Hamiltonian, split $(0, t)$ into N intervals of width $\Delta = t/N$. Then

$$\begin{aligned} |\psi(t)\rangle &= |\psi(0)\rangle + \Delta \left. \frac{d|\psi\rangle}{dt} \right|_0 = |\psi(0)\rangle - \frac{i\Delta}{\hbar} H(0) |\psi(0)\rangle \\ &= \left[1 - \frac{i\Delta}{\hbar} H(0) \right] |\psi(0)\rangle \\ &\approx \exp \left[-\frac{i\Delta}{\hbar} H(0) \right] |\psi(0)\rangle \end{aligned}$$

The overall propagator is found by summing the changes:

$$\begin{aligned} |\psi(t)\rangle &= \prod_{n=0}^{N-1} e^{-i\Delta H(n\Delta)/\hbar} |\psi(0)\rangle \\ U(t) &= \exp \left[-\frac{i}{\hbar} \int_0^t H(t') dt' \right], \quad N \rightarrow \infty \end{aligned} \tag{2}$$

4 Simple Problems in 1D

4.1 The Free Particle

4.1.1 Allowed Energy States

Start from the Schrodinger equation:

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle = \frac{P^2}{2m} |\psi\rangle$$

The normal modes (in energy basis) are

$$|\psi\rangle = e^{-iEt/\hbar} |E\rangle$$

We seek to find allowed E states. $|p\rangle$ is an eigenvector of both P and P^2 , so

$$\begin{aligned} \frac{P^2}{2m} |p\rangle = E |p\rangle &\longrightarrow \frac{p^2}{2m} |p\rangle = E |p\rangle \\ p &= \pm\sqrt{2mE} \end{aligned}$$

Thus for each eigenvalue E , we have 2 eigenvectors:

$$\begin{aligned} |E, +\rangle &= |p = \sqrt{2mE}\rangle \\ |E, -\rangle &= |p = -\sqrt{2mE}\rangle \end{aligned}$$

I.e. a particle with energy E may be found moving left or right with momentum $\sqrt{2mE}$. So far, this agrees with a classical analysis. The difference is that in QM, we can have a state like

$$|E\rangle = \beta |p = \sqrt{2mE}\rangle + \gamma |p = -\sqrt{2mE}\rangle$$

4.1.2 Propagator

Use the propagator from the time-independent Schrodinger equation:

$$U(t) = \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-iE(p)t/\hbar} dp = \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-ip^2t/2m\hbar} dp$$

Note that in the X basis, the propagator for a time slice is

$$\begin{aligned} U(x, t; x') &\equiv \langle x|U(t)|x'\rangle = \int_{-\infty}^{\infty} \langle x|p\rangle \langle p|x'\rangle e^{-ip^2t/2m\hbar} dp \\ &= \frac{1}{2m\hbar} \int_{-\infty}^{\infty} e^{ip(x-x')/\hbar} \cdot e^{-ip^2t/2m\hbar} dp \\ &= \sqrt{\frac{m}{2\pi\hbar it}} e^{im(x-x')^2/2\hbar t} \end{aligned}$$

In terms of the propagator, given the initial state $\psi(x', 0)$, we can find an arbitrary state $\psi(x, t)$ as

$$\psi(x, t) = \int U(x, t; x') \psi(x', 0) dx'$$

4.2 The Particle in a Box

4.2.1 Wavefunction

Consider a box potential:

$$V(x) = \begin{cases} 0 & |x| < L/2 \\ \infty & |x| \geq L/2 \end{cases}$$

Let I and III be the regions outside the box, while II is the region within the box. Use the eigenvalue equation in the X basis:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

In III, we have

$$\begin{aligned} \frac{d^2\psi_{\text{III}}}{dx^2} - \frac{2m(V_0 - E)}{\hbar^2}\psi_{\text{III}} &= 0 \quad , \quad V_0 \rightarrow \infty \\ \psi_{\text{III}} &= Ae^{-\kappa x} + Be^{\kappa x} \quad , \quad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \end{aligned}$$

We must have $B = 0$ because otherwise the wavefunction blows up as $x \rightarrow \infty$. Then as $V_0 \rightarrow \infty$, $\psi_{\text{III}} = 0$. This is as expected. Similar results hold for I: $\psi_{\text{I}} = 0$.

In II, the solution is that of a free particle:

$$\psi_{\text{II}} = Ae^{ikx} + Be^{-ikx} \quad , \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

Further, it must obey the boundary conditions:

$$\begin{aligned} \psi_{\text{I}}(-L/2) = \psi_{\text{II}}(-L/2) = 0 &\longrightarrow Ae^{ikL/2} + Be^{ikL/2} = 0 \\ \psi_{\text{III}}(L/2) = \psi_{\text{II}}(L/2) = 0 &\longrightarrow Ae^{ikL/2} + Be^{-ikL/2} = 0 \\ \begin{bmatrix} e^{-ikL/2} & e^{ikL/2} \\ e^{ikL/2} & e^{-ikL/2} \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

For this to happen, the determinant must be 0, i.e. $e^{ikL} - e^{-ikL} = 0$, thus $k = n\pi/L$, where $n = 0, \pm 1, \pm 2, \dots$. We can find that $A = -e^{in\pi}B$, leading to two families of solutions:

$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) & n \text{ even} \\ \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi x}{L}\right) & n \text{ odd} \end{cases}$$

4.2.2 Allowed Energies

Note that the energy is quantized:

$$E = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

And further, the lowest bound on energy (for non-trivial solutions) is nonzero: the ground state energy E_g is

$$E_g = \frac{\hbar^2 \pi^2}{2mL^2}$$

An “intuitive” explanation for this is that Δx is bounded in the well, so by the uncertainty principle, we cannot have a well-defined momentum of 0.

The uncertainty principle also provides a quick estimate of the ground energy. Start with $H = P^2/2m$. From the Ehrenfest theorem, $\langle H \rangle = \langle P^2 \rangle / 2m$. Since $\langle P \rangle = 0$ inside the box, we can further write

$$\langle H \rangle = \frac{\langle (P - \langle P \rangle)^2 \rangle}{2m} = \frac{(\Delta P)^2}{2m}$$

From the uncertainty relations, $\Delta X \cdot \Delta P \geq \hbar/2$, so we have

$$\langle H \rangle \geq \frac{\hbar^2}{8m(\Delta x)^2} \geq \frac{\hbar^2}{2mL^2}$$

In energy eigenstates, $\langle H \rangle = E$. Thus we have

$$E \geq \frac{\hbar^2}{2mL^2}$$

Which is the estimated order of the ground state, accurate up to a factor of π^2 .

4.3 The Continuity Equation of Probability

From classical electrodynamics:

$$\text{(global)} \quad Q_{\text{total}} = \text{const.}$$

$$\text{(local)} \quad \frac{d}{dt} \int_V \rho(\mathbf{r}, t) d^3\mathbf{r} = - \int_{S_V} \mathbf{j} \cdot d\mathbf{S}$$

In QM:

$$\text{(global)} \quad \iiint P(\mathbf{r}, t) d^3\mathbf{r} = \text{const.}$$

$$\text{(local)} \quad \frac{d}{dt} \int P(\mathbf{r}, t) d^3\mathbf{r} = - \int_{S_\infty} \mathbf{j} \cdot d\mathbf{S} \quad , \quad \mathbf{j} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

In a nutshell, the total probability to find a particle anywhere in the universe is conserved. Locally, the reduction in probability of a region must correspond to a change in the probability current $\mathbf{j}$ is the **probability current density**.

4.4 Quantum Tunneling

The details are complicated, see Shankar chp. 5 for more. The main idea: instead of an infinite potential, we have a region of 0 potential I, and a region of *finite* potential II. Let $V = 0$ in I and $V = V_0$ in II.

$$\begin{aligned} \text{I: } \psi_E(x) &= Ae^{ik_1x} + Be^{-ik_1x} \quad , \quad k_1 = \sqrt{\frac{2mE}{\hbar^2}} \\ \text{II: } \psi_E(x) &= Ce^{ik_2x} + De^{-ik_2x} \quad , \quad k_2 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \end{aligned}$$

Set $D = 0$ for the right-traveling transmitted waves. At $x = 0$, boundary conditions require that

$$\begin{aligned} A + B &= C \\ ik_1(A - B) &= ik_2C \end{aligned}$$

After some calculations, we find the reflection probability (R) and transmission probability (T) to be:

$$\begin{aligned} R &= \left| \frac{B}{A} \right|^2 = \left| \frac{\sqrt{E_0} - \sqrt{E_0 - V_0}}{\sqrt{E_0} + \sqrt{E_0 - V_0}} \right|^2 \\ T &= 1 - R = \left| \frac{4\sqrt{E_0}\sqrt{E_0 - V_0}}{(\sqrt{E_0} + \sqrt{E_0 - V_0})^2} \right| \end{aligned}$$

The key is that even if $E_0 < V_0$ (energy not enough to overcome barrier), there's still a nonzero probability for the particle to be transmitted. This is called **quantum tunneling**.

4.5 Ehrenfest's Theorem

QM applied to many particles should reproduce the macroscopic world.

Consider the time evolution of expectation values:

$$\frac{d}{dt} \langle \Omega \rangle = \frac{d}{dt} \langle \psi | \Omega | \psi \rangle = \langle \dot{\psi} | \Omega | \psi \rangle + \langle \psi | \Omega | \dot{\psi} \rangle + \langle \psi | \dot{\Omega} | \psi \rangle$$

Assume Ω has no time dependence, so $\dot{\Omega} = 0$. Further, from the Schrodinger equation,

$$\begin{aligned} |\dot{\psi}\rangle &= -\frac{i}{\hbar} H |\psi\rangle \\ \langle \dot{\psi}| &= \frac{i}{\hbar} \langle \psi| H \end{aligned}$$

Combining the two results gives **Ehrenfest's theorem**:

$$\frac{d}{dt} \langle \Omega \rangle = -\frac{i}{\hbar} \langle [\Omega, H] \rangle$$

Note the similarity with classical mechanics:

$$\frac{d\omega}{dt} = \{\omega, H\}$$

For a Hamiltonian of the form $H = P^2/2m + V(X)$, we can also get similar forms to the canonical equations:

$$\langle \dot{X} \rangle = \left\langle \frac{\partial H}{\partial P} \right\rangle \quad , \quad \langle \dot{P} \rangle = \left\langle -\frac{\partial H}{\partial X} \right\rangle$$

Thus we can regain classical physics if the fluctuation about the mean is small. But the above equations depend on the choice of H .

5 The Harmonic Oscillator

5.1 The Classical Oscillator

Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

Equations of motion:

$$\begin{aligned} \dot{x} &= \frac{\partial H}{\partial p} = \frac{p}{m} \\ \dot{p} &= -\frac{\partial H}{\partial x} = -m\omega^2 x \end{aligned}$$

From here, the overall motion is apparent and can be solved:

$$\ddot{x} + \omega^2 x = 0$$

5.2 Quantum Oscillator in the X Basis

To quantize the oscillator, we treat it as a particle $|\psi\rangle$ obeying the Schrodinger equation.

$$\begin{aligned} i\hbar \frac{d}{dt} |\psi\rangle &= H |\psi\rangle \\ H &= \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2 \end{aligned}$$

Solving the eigenvalue equation for energy eigenstates,

$$\begin{aligned} H |E\rangle &= E |E\rangle \\ \left(\frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2 \right) |E\rangle &= E |E\rangle \end{aligned}$$

In the X basis, we make the substitutions $P \rightarrow -i\hbar \frac{d}{dx}$, $X \rightarrow x$, and $|E\rangle \rightarrow \psi_E$. Then

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} \left(E - \frac{1}{2}m\omega^2 x^2 \right) \psi = 0$$

To solve this, we substitute $x = by$ and $\varepsilon = E/\hbar\omega$, where $b = \sqrt{\hbar/m\omega}$. Then $d^2\psi/dy^2 + (2\varepsilon - y^2)\psi = 0$, which has solution $\psi(y) = u(y)e^{-y^2/2}$, where $u(y)$ goes to $A = cy$ as $y \rightarrow 0$ and y^m as $y \rightarrow \infty$. By requiring that the solution be in Hilbert space (normalizable to unity or the delta function), $\varepsilon = (2n + 1)/2$, and the energy is quantized:

$$E_n = \varepsilon_n \hbar\omega = \left(n + \frac{1}{2}\right) \hbar\omega$$

We can find the explicit wavefunction as $\psi_E(y) = AH_n(y)e^{-y^2/2}$, where A is a normalization factor and H_n is a Hermite polynomial of degree n .

$$\psi_n(x) = \underbrace{\left(\frac{m\omega}{\pi\hbar e^{2n}(n!)^2}\right)^{1/4}}_A \cdot H_n \left[\sqrt{\frac{m\omega}{\hbar}} x \right] \cdot \exp\left(-\frac{m\omega x^2}{2\hbar}\right)$$

The propagator can be found as

$$\begin{aligned} U(x, t; x', t') &= \sum_{n=0}^{\infty} |\psi_n\rangle \langle \psi_n| e^{-iEt/\hbar} \\ &= \sum_{n=0}^{\infty} A_n^2 \exp\left(-\frac{m\omega x^2}{2\hbar}\right)^2 H_n(x) H_n(x') \exp\left(-i\left(n + \frac{1}{2}\right)(t - t')\omega\right) \end{aligned}$$

5.3 Quantum Oscillator in the Energy Basis

A slightly easier method is to use the commutator relation $[X, P] = i\hbar$, which is basis-independent, in the energy basis. Define the operators:

$$\begin{aligned} a &= \sqrt{\frac{m\omega}{2\hbar}} X + i\sqrt{\frac{1}{2m\omega\hbar}} P \\ a^\dagger &= \sqrt{\frac{m\omega}{2\hbar}} X - i\sqrt{\frac{1}{2m\omega\hbar}} P \\ [a, a^\dagger] &= 1 \end{aligned}$$

So that the Hamiltonian $H = P^2/2m + m\omega^2 X^2/2$ becomes

$$H = \left(a^\dagger a + \frac{1}{2}\right) \hbar\omega$$

Let $\hat{H} = H/\hbar\omega$. Using a and $a^\dagger$, an eigenstate $|n\rangle$ of $\hat{H}$ can be used to generate another eigenstate of $\hat{H}$.

$$\begin{aligned} \hat{H}a|n\rangle &= (n-1)a|n\rangle \longrightarrow a|n\rangle = \sqrt{n}|n-1\rangle \\ \hat{H}a^\dagger|n\rangle &= (n+1)a^\dagger|n\rangle \longrightarrow a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \end{aligned}$$

with appropriate normalization. a and $a^\dagger$ are called the lowering/raising operators, or destruction/creation operators. The number operator is $H = a^\dagger a$, such that $N|n\rangle = n|n\rangle$. The vacuum state is easily identified as $a|n_0\rangle = 0$, or

$$n_0 = \frac{1}{2}, \quad E = n_0 \hbar \omega = \frac{1}{2} \hbar \omega$$

For example, to calculate $\langle 3|X^3|2\rangle$,

$$\langle 3|X^3|2\rangle = \left(\frac{\hbar}{2m\omega}\right)^{3/2} \langle 3|(a + a^\dagger)^3|2\rangle$$

After expanding and only taking contributions from $a^\dagger a^\dagger a$, $a^\dagger a a^\dagger$, and $a a^\dagger a^\dagger$, we have

$$\langle 3|X^3|2\rangle = 9\sqrt{3} \left(\frac{\hbar}{2m\omega}\right)^{3/2}$$

6 The Path Integral Formulation

6.1 Overview

The Schrodinger formalism uses the Hamiltonian, while the path integral formalism (from Feynman) uses the Lagrangian. The path integral formalism gives the full propagator more easily.

To find $U(x, t; x', t')$,

1. Draw all paths from (x', t') to (x, t) .
2. Find the action $S[x(t)]$ for each path.
3. $U(x, t; x', t') = A \sum e^{iS[x(t)]/\hbar}$

6.2 The Free Particle

6.2.1 Classical Approximation

Note that paths can destructively interfere. Using just the classical path is an okay approximation for macroscopic objects.

$$\begin{aligned} x_{\text{cl}}(t'') &= x' + \frac{x - x'}{t - t'}(t'' - t') \\ L &= T - V = \frac{1}{2}m \left(\frac{x - x'}{t - t'}\right)^2 = \text{const.} \\ S_{\text{cl}} &= \int_{t'}^t L dt'' = \frac{1}{2}m \left(\frac{x - x'}{t - t'}\right)^2 (t - t') \end{aligned}$$

The propagator:

$$U(t) = A' e^{iS_{\text{cl}}/\hbar} \approx A' \exp\left(\frac{im(x - x')^2}{2\hbar(t - t')}\right)$$

To find A' , the normalization constant, we require $U \rightarrow \delta(x - x')$ as $(t - t') \rightarrow 0$. Recalling the Gaussian definition of the delta function,

$$\delta(x - x') = \lim_{\Delta \rightarrow 0} \frac{1}{\sqrt{\pi\Delta^2}} \exp\left(-\frac{(x - x')^2}{\Delta^2}\right) \quad , \quad \Delta = \sqrt{\frac{2i\hbar(t - t')}{m}}$$

$$A' = \sqrt{\frac{m}{2\pi\hbar i(t - t')}}}$$

Thus the propagator, to a classical approximation, is

$$U(x, t; x', t') \equiv U(x, t; x') = \sqrt{\frac{m}{2\pi\hbar i t}} \exp\left(\frac{im(x - x')^2}{2\hbar t}\right)$$

Which agrees with the one found classically, through the Schrodinger equation.

6.2.2 Full Solution

Without approximations, the solution is a little harder. Let $(x_N; t_N)$ be the endpoint and $(x_0; t_0)$ be the starting point. The path integral is

$$U(x_N, t_N; x_0, t_0) = \int_{x_0}^{x_N} e^{iS[x(t)]/\hbar} \mathcal{D}[x(t)]$$

$\mathcal{D}[x(t)]$ means “integrate over all paths connecting x_0 and x_N in the interval (t_0, t_N) .” Approximate by splitting the path into $(N + 1)$ intervals $x_0, x_1, \dots, x_N$ corresponding to $t_0, t_1, \dots, t_N$, where $t_n = t_0 + n\varepsilon$, with $\varepsilon = (t_N - t_0)/N$. Interpolate with straight lines. Then set $N \rightarrow \infty$ at the end.

Thus

$$S = \int_{t_0}^{t_N} L dt \rightarrow \sum_{i=0}^{N-1} \frac{1}{2} m \left(\frac{x_{i+1} - x_i}{\varepsilon} \right)^2 \varepsilon$$

$$U(x_N, t_N; x_0, t_0) = \int_{x_0}^{x_N} e^{iS[x(t)]/\hbar} \mathcal{D}[x(t)] = \lim_{N \rightarrow \infty} A \int \int \cdots \int \exp \left[\frac{im}{2\hbar} \sum_{i=0}^{N-1} \frac{(x_{i+1} - x_i)^2}{\varepsilon} \right] dx_1 dx_2 \cdots dx_{N-1}$$

By substituting $y_i = \sqrt{m/2\hbar\varepsilon} x_i$,

$$U(x_N, t_N; x_0, t_0) = A \left(\frac{2\hbar\varepsilon}{m} \right)^{(N-1)/2} \lim_{N \rightarrow \infty} \int \int \cdots \int \exp \left[- \sum_{i=0}^{N-1} \frac{(y_{i+1} - y_i)^2}{\varepsilon} \right] dy_1 dy_2 \cdots dy_{N-1}$$

By finding the integral portion of dy_1 , then dy_2 , and so on, the total integral is found to be

$$\sqrt{\frac{(i\pi)^{N-1}}{N}} e^{-(y_N - y_0)^2 / iN}$$

Thus the propagator is

$$\begin{aligned} U(x_N, t_N; x_0, t_0) &= A \left(\frac{2\hbar\varepsilon}{m} \right)^{(N-1)/2} \cdot (i\pi)^{(N-1)/2} \cdot \frac{1}{\sqrt{N}} \cdot \exp \left(\frac{im(x_N - x_0)^2}{2\hbar N\varepsilon} \right) \\ &= A \left(\frac{2\pi\hbar\varepsilon i}{m} \right)^{N/2} \left(\frac{m}{2\pi\hbar i N\varepsilon} \right)^{1/2} \exp \left(\frac{im(x_N - x_0)^2}{2\hbar N\varepsilon} \right) \end{aligned}$$

As $N \rightarrow \infty$ and $N\varepsilon \rightarrow t_N - t_0 \equiv t$, we recover the full propagator given a specific value of A , which agrees with the ones before.

$$\begin{aligned} A &= \left(\frac{2\pi\hbar\varepsilon i}{m} \right)^{-N/2} \\ U(x_N, t_N; x_0, t_0) &= \sqrt{\frac{m}{2\pi\hbar i t}} \exp \left(\frac{im(x_N - x_0)^2}{2\hbar t} \right) \end{aligned}$$

By convention, let $A \equiv B^{-N}$, such that

$$\begin{aligned} B &= \sqrt{\frac{2\pi\hbar\varepsilon i}{m}} \\ \int \mathcal{D}[x(t)] &= \lim_{N \rightarrow \infty} \frac{1}{B} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \frac{dx_1}{B} \frac{dx_2}{B} \dots \frac{dx_{N-1}}{B} \end{aligned}$$

7 The Heisenberg Uncertainty Relations

7.1 Derivation

Let Ω, Λ be Hermitian and $[\Omega, \Lambda] = i\Gamma$. Also define $\hat{\Omega} = \Omega - \langle \Omega \rangle$ and $\hat{\Lambda} = \Lambda - \langle \Lambda \rangle$.

$$\begin{aligned} (\Delta\Omega)^2(\Delta\Lambda)^2 &= \langle \psi | (\Omega - \langle \Omega \rangle)^2 | \psi \rangle \langle \psi | (\Lambda - \langle \Lambda \rangle)^2 | \psi \rangle \\ &= \langle \psi | \hat{\Omega}^2 | \psi \rangle \langle \psi | \hat{\Lambda}^2 | \psi \rangle \\ &= \langle \hat{\Omega}\psi | \hat{\Omega}\psi \rangle \langle \hat{\Lambda}\psi | \hat{\Lambda}\psi \rangle \end{aligned}$$

Recall the Schwartz inequality $|V_1|^2|V_2|^2 \geq |\langle V_1 | V_2 \rangle|^2$. Thus

$$\begin{aligned} (\Delta\Omega)^2(\Delta\Lambda)^2 &\geq |\langle \hat{\Omega}\psi | \hat{\Lambda}\psi \rangle|^2 = |\langle \psi | \hat{\Omega}^\dagger \hat{\Lambda} | \psi \rangle|^2 \\ &= |\langle \psi | \hat{\Omega} \hat{\Lambda} | \psi \rangle|^2 \\ &= \left| \langle \psi | \frac{\{\hat{\Omega}, \hat{\Lambda}\}}{2} + \frac{[\hat{\Omega}, \hat{\Lambda}]}{2} | \psi \rangle \right|^2 \\ &= \frac{1}{4} \left| \langle \psi | \{\hat{\Omega}, \hat{\Lambda}\} | \psi \rangle + i \langle \psi | \Gamma | \psi \rangle \right|^2 \\ &= \frac{1}{4} \langle \psi | \{\hat{\Omega}, \hat{\Lambda}\} | \psi \rangle^2 + \frac{1}{4} \langle \psi | \Gamma | \psi \rangle^2 \end{aligned}$$

For the canonically conjugate operators like X and P , $\Gamma = \hbar$, so

$$\begin{aligned} (\Delta\Omega)^2(\Delta\Lambda)^2 &\geq \frac{1}{4} \langle\psi|\{\hat{\Omega}, \hat{\Lambda}\}|\psi\rangle^2 + \frac{1}{4}\hbar^2 \\ \Delta\Omega \cdot \Delta\Lambda &\geq \hbar/2 \end{aligned}$$

7.2 Equality

Schwartz inequality is an equality iff. $|V_1\rangle = c|V_2\rangle$. So the minimum case occurs when

$$\begin{aligned} \hat{\Omega}|\psi\rangle &= c\hat{\Lambda}|\psi\rangle \\ \langle\psi|\{\hat{\Omega}, \hat{\Lambda}\}|\psi\rangle & \end{aligned}$$

For X, P , this becomes

$$\begin{aligned} (P - \langle P \rangle)|\psi\rangle &= c(X - \langle X \rangle)|\psi\rangle \\ \langle\psi|(P - \langle P \rangle)(X - \langle X \rangle) &+ (X - \langle X \rangle)(P - \langle P \rangle)|\psi\rangle \end{aligned}$$

Project onto the X basis and shift frames so that $\langle X \rangle = 0$ for now. We get

$$\begin{aligned} \left(-i\hbar\frac{d}{dx} - \langle P \rangle\right)\psi(x) &= cx\psi(x) \quad \longrightarrow \quad \psi(x) = \psi(0)e^{i\langle P \rangle x/\hbar}e^{icx^2/2\hbar} \\ \langle\psi|c^*X^2 + cX^2|\psi\rangle &= 0 \quad \longrightarrow \quad (c + c^*)\langle\psi|X^2|\psi\rangle = 0 \end{aligned}$$

So c must be imaginary, $c = i|c|$. Let $\Delta^2 = \hbar/|c|$. Then

$$\psi(x) = \psi(0)e^{i\langle P \rangle x/\hbar}e^{-x^2/2\Delta^2}$$

For arbitrary X with $\langle X \rangle \neq 0$, we have

$$\psi(x) = \psi(\langle X \rangle)e^{i\langle P \rangle(x - \langle X \rangle)/\hbar}e^{-(x - \langle X \rangle)^2/2\Delta^2}$$

I.e. a Gaussian function of arbitrary width provides the lower-bound solution for the Heisenberg uncertainty relation.

7.3 Energy-Time Relation

A lesser-known but important relation:

$$\Delta E \cdot \Delta t \geq \hbar/2$$

One interpretation: violations in energy conservation by ΔE are possible over times $\Delta t \sim \hbar/\Delta E$.

E.g. nucleons are held together by an exchange of pions. A nucleon can produce pions out of thin air, as long as it's caught by the receiver within $\Delta t \sim \hbar/\Delta E = \hbar/\mu c^2$. If the pion travels at c , it can travel at most $r = c\Delta t = \hbar/\mu c$, the **Compton wavelength**. This is the maximal range of the nuclear force.

8 Systems with N Degrees of Freedom

8.1 The Direct Product Space $\mathbb{V}_{1\otimes 2}$

Let system 1 have $|x_1\rangle, |p_1\rangle$ in $\mathbb{V}_1$ and system 2 have $|x_2\rangle, |p_2\rangle$ in $\mathbb{V}_2$. The **direct product** (or **tensor product**) $|x_1\rangle \otimes |x_2\rangle$ is in $\mathbb{V}_{1\otimes 2}$.

- **Linear:** $(\alpha_1 |x_1\rangle + \alpha_2 |x_2\rangle) \otimes \beta |x_3\rangle = \alpha_1 \beta |x_1\rangle \otimes |x_3\rangle + \alpha_2 \beta |x_2\rangle \otimes |x_3\rangle$
- **Inner product:** $(\langle x'_1| \otimes \langle x'_2|) (|x_1\rangle \otimes |x_2\rangle) = \langle x'_1|x_1\rangle \langle x'_2|x_2\rangle = \delta(x'_1 - x_1)\delta(x'_2 - x_2)$.

Operators on either $\mathbb{V}_1$ or $\mathbb{V}_2$ can be promoted to operators on $\mathbb{V}_{1\otimes 2}$, e.g.

$$\begin{aligned} X_1^{(1)} &\longrightarrow X_1^{(1)} \otimes I^{(2)} \equiv X_1^{(1)\otimes(2)} \\ P_2^{(2)} &\longrightarrow I^{(1)} \otimes P_2^{(2)} \equiv P_2^{(1)\otimes(2)} \end{aligned}$$

This obeys the following relation:

$$(\Omega_1^{(1)} \otimes \Gamma_2^{(2)}) (\Theta_1^{(1)} \otimes \Lambda_2^{(2)}) = (\Omega\Theta)_1^{(1)} \otimes (\Gamma\Lambda)_2^{(2)}$$

The commutation relations are:

$$\begin{aligned} [X_i^{(1)\otimes(2)}, P_j^{(1)\otimes(2)}] &= i\hbar \delta_{ij} I^{(1)\otimes(2)} \\ [X_i^{(1)\otimes(2)}, X_j^{(1)\otimes(2)}] &= [P_i^{(1)\otimes(2)}, P_j^{(1)\otimes(2)}] = 0 \end{aligned}$$

From now on, notation will be simplified:

$$\begin{aligned} |x_1\rangle \otimes |x_2\rangle &\longrightarrow |x_1 x_2\rangle \\ X_1^{(1)\otimes(2)} &\longrightarrow X_1 \end{aligned}$$

8.2 The Two-Particle State

8.2.1 Separable Hamiltonians

We know $H = H_1 + H_2$.

In the energy basis, let $|E\rangle = |E_1\rangle \otimes |E_2\rangle$ and $E = E_1 + E_2$.

$$\begin{aligned} H |E\rangle &= (E_1 + E_2) |E_1\rangle \otimes |E_2\rangle = E |E\rangle \\ |\psi(t)\rangle &= e^{-iEt/\hbar} |E\rangle = e^{-iE_1 t/\hbar} |E_1\rangle \otimes e^{-iE_2 t/\hbar} |E_2\rangle \end{aligned}$$

We can also use the explicit X basis and substitute for P and X .

$$\begin{aligned} H |E\rangle &= E |E\rangle \\ \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x_1^2} + V(x_1) - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x_2^2} + V(x_2) \right] \psi_E(x_1, x_2) &= E \psi_E(x_1, x_2) \end{aligned}$$

If the states are separable, $\psi_E(x_1, x_2) = \psi_{E_1}(x_1)\psi_{E_2}(x_2)$. By plugging in and solving, we also get

$$\psi_E(x_1, x_2) = e^{-iE_1 t/\hbar} \psi_{E_1}(x_1) \cdot e^{-iE_2 t/\hbar} \psi_{E_2}(x_2)$$

Alternatively, if the potential is of the form $V(x_1, x_2) = V(x_1 - x_2)$, the problem can be solved. Consider:

$$H = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(x_1, x_2)$$

$$V(x_1, x_2) \neq V_1(x_1) + V_2(x_2)$$

Define the following coordinates:

$$x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$x_{\text{rel}} = x_1 - x_2$$

The Hamiltonian then reduces to

$$H = H_{\text{CM}} + H_{\text{rel}} = \frac{p_{\text{CM}}^2}{2M} + \frac{p_{\text{rel}}^2}{2\mu} + V(x_{\text{rel}})$$

$$M = m_1 + m_2, \quad p_{\text{CM}} = M \dot{x}_{\text{CM}}$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \quad p = \mu \dot{x}_{\text{rel}}$$

The eigenfunctions of H factorize:

$$E = \frac{p_{\text{CM}}^2}{2M} + E_{\text{rel}}$$

$$\psi_E(x_1, x_2) = \psi_{E_{\text{CM}}}(x_{\text{CM}}) \cdot \psi_{E_{\text{rel}}}(x_{\text{rel}}) = \frac{e^{ip_{\text{CM}} \cdot x_{\text{CM}}/\hbar}}{\sqrt{2\pi\hbar}} \cdot \psi_{E_{\text{rel}}}(x_{\text{rel}})$$

By studying the system in the CM frame, we can further simplify to $E \rightarrow E_{\text{rel}}, \psi_E \rightarrow C\psi_{E_{\text{rel}}}(x_{\text{rel}})$.

8.3 Identical Particles

8.3.1 Bosons and Fermions

Unlike classical mechanics, in QM, we don't know the exact history of a system. The following situations:

- Particle a at x_1 , particle b at x_2
- Particle b at x_1 , particle a at x_2

are indistinguishable if a and b are identical.

We do know that they should be eigenvectors of $X_1 + X_2$ with eigenvalue $a + b$. In the eigenstate of $X_1 + X_2$, there are 2 options:

1. Symmetric: $|\psi\rangle = \frac{1}{\sqrt{2}} (|ab\rangle + |ba\rangle) \equiv |ab, S\rangle$
 - **Bosons:** pion, photon, graviton
 - Spin is integer multiple of $\hbar$

2. Antisymmetric: $|\psi\rangle = \frac{1}{\sqrt{2}}(|ab\rangle - |ba\rangle) \equiv |ab, A\rangle$
- **Fermions:** electron, proton, neutron
 - Spin is half-integer multiple of $\hbar$

From this we get the **Pauli exclusion principle:** 2 fermions can't be in the same quantum state. If they were, then

$$|\omega_1\omega_2, A\rangle = |\omega_1\omega_2\rangle - |\omega_2\omega_1\rangle \equiv |\omega_1\omega_1\rangle - |\omega_1\omega_1\rangle = 0$$

8.3.2 Spin Statistics

The probability to find half a symmetric/antisymmetric particle pair in x_1 or x_2 , given energy measurements of $n_1 = 3$ and $n_2 = 4$:

$$\begin{aligned} P_{S/A}(x_1, x_2) &= 2 \left| \psi_{S/A}(x_1, x_2) \right|^2 = 2 \left| \frac{1}{\sqrt{2}} [\psi_3(x_1)\psi_4(x_2) \pm \psi_4(x_1)\psi_3(x_2)] \right|^2 \\ &= |\psi_3(x_1)|^2 |\psi_4(x_2)|^2 + |\psi_4(x_1)|^2 |\psi_3(x_2)|^2 \\ &\quad \pm \left[\psi_3^*(x_1)\psi_4(x_1)\psi_4^*(x_2)\psi_3(x_2) + \psi_4^*(x_1)\psi_3(x_1)\psi_3^*(x_2)\psi_4(x_2) \right] \end{aligned}$$

The 3rd term (**interference term**) shows that quantum statistics is different from classical statistics. There's something more than just summing both probabilities of $(x_1, 3; x_2, 4)$ or $(x_2, 3; x_1, 4)$.

If $x_1 = x_2 = x$, then

- Bosons tend to **clump together**. Bose-Einstein statistics. Note that the probability is twice that of the classical probability to find either $(x_1, 3; x_2, 4)$ or $(x_1, 4; x_2, 3)$.

$$P_S = 2 \left[|\psi_3(x_1)|^2 |\psi_4(x_2)|^2 + |\psi_4(x_1)|^2 |\psi_3(x_2)|^2 \right]$$

- Fermions tend to **avoid each other**. Fermi-Dirac statistics.

$$P_A = 0$$

8.3.3 Symmetrization/Antisymmetrization of N Particles

Assume $n = 3$ and n_1, n_2, n_3 all distinct. To find a symmetric state, use the **totally symmetric state**.

$$\begin{aligned} |n_1 n_2 n_3, S\rangle &= \frac{1}{\sqrt{3!}} [|n_1 n_2 n_3\rangle + |n_1 n_3 n_2\rangle + |n_2 n_1 n_3\rangle \\ &\quad + |n_2 n_3 n_1\rangle + |n_3 n_1 n_2\rangle + |n_3 n_2 n_1\rangle] \end{aligned}$$

For an antisymmetric state, use the **totally antisymmetric state**. Signs depend on the Levi-Civita symbol of $\epsilon_{n_1 n_2 n_3}$.

$$\begin{aligned} |n_1 n_2 n_3, A\rangle &= \frac{1}{\sqrt{3!}} [|n_1 n_2 n_3\rangle - |n_1 n_3 n_2\rangle + |n_2 n_3 n_1\rangle \\ &\quad - |n_2 n_1 n_3\rangle + |n_3 n_1 n_2\rangle - |n_3 n_2 n_1\rangle] \end{aligned}$$

Alternatively, use a **Slater determinant**. This also makes it easier to see that the fermion wavefunction vanishes if multiple x 's or n 's coincide.

$$\psi_{n_1 n_2 n_3}(x_1, x_2, x_3, A) = \frac{1}{\sqrt{3!}} \begin{vmatrix} \psi_{n_1}(x_1) & \psi_{n_2}(x_1) & \psi_{n_3}(x_1) \\ \psi_{n_1}(x_2) & \psi_{n_2}(x_2) & \psi_{n_3}(x_2) \\ \psi_{n_1}(x_3) & \psi_{n_2}(x_3) & \psi_{n_3}(x_3) \end{vmatrix}$$

9 Transformations and Their Generators

Classical mechanics:

- Translation: $x \rightarrow x + \varepsilon$ and $p \rightarrow p$
- Translational invariance: $H \rightarrow H$
- Conservation law: $\dot{p} = 0$

Quantum mechanics:

- Translation: $\langle X \rangle \rightarrow \langle X \rangle + \varepsilon$ and $\langle P \rangle \rightarrow \langle P \rangle$
- Translational invariance: $\langle H \rangle \rightarrow \langle H \rangle$
- Conservation law: $\langle \dot{P} \rangle = 0$

9.1 Translations

9.1.1 Active vs. Passive Transformations

Under an infinitesimal translation, each state $|\psi\rangle$ goes to $|\psi_\varepsilon\rangle$ such that

$$\begin{aligned} \langle \psi_\varepsilon | X | \psi_\varepsilon \rangle &= \langle \psi | X | \psi \rangle + \varepsilon \\ \langle \psi_\varepsilon | P | \psi_\varepsilon \rangle &= \langle \psi | P | \psi \rangle \end{aligned}$$

In the **active transformation picture**, define a translation operator $T(\varepsilon)$ such that $T(\varepsilon) |\psi\rangle = |\psi_\varepsilon\rangle$. This corresponds to actually moving the particle by ε .

$$\begin{aligned} \langle \psi_\varepsilon | T^\dagger(\varepsilon) X T(\varepsilon) | \psi_\varepsilon \rangle &= \langle \psi | X | \psi \rangle + \varepsilon \\ \langle \psi_\varepsilon | T^\dagger(\varepsilon) P T(\varepsilon) | \psi_\varepsilon \rangle &= \langle \psi | P | \psi \rangle \end{aligned}$$

In the **passive translation picture**, the operators X and P themselves are replaced. This corresponds to shifting the frame of reference by ε .

$$\begin{aligned} X &\rightarrow T^\dagger(\varepsilon) X T(\varepsilon) = X + \varepsilon I \\ P &\rightarrow T^\dagger(\varepsilon) P T(\varepsilon) = P \end{aligned}$$

For definite position kets $|x\rangle$, we have $T(\varepsilon)|x\rangle = |x + \varepsilon\rangle$. Thus for a general $|\psi\rangle$,

$$\begin{aligned} T(\varepsilon)|\psi\rangle &= T(\varepsilon) \int_{-\infty}^{\infty} |x\rangle \langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty} \langle x + \varepsilon\rangle \langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty} \langle x'\rangle \langle x' - \varepsilon|\psi\rangle dx' \end{aligned}$$

Thus

$$\langle x|T(\varepsilon)|\psi\rangle = \psi(x - \varepsilon)$$

E.g. if $\psi(x) = \langle x|\psi\rangle \sim e^{-x^2}$, then $\psi_\varepsilon(x) \sim e^{-(x-\varepsilon)^2}$.

9.1.2 Momentum as a Generator of Translations

First, find the explicit form of $T(\varepsilon)$. Use $T(0) = I$. Assume the form

$$T(\varepsilon) = I - \frac{i\varepsilon}{\hbar} G$$

call G the **generator** of the transformation. Then

$$\begin{aligned} \langle x|T(\varepsilon)|\psi\rangle &= \psi(x - \varepsilon) \\ \langle x|\psi\rangle - \frac{i\varepsilon}{\hbar} \langle x|G|\psi\rangle &\approx \psi(x) - \frac{d\psi}{dx}\varepsilon \\ \langle x|G|\psi\rangle &\approx -i\hbar \frac{d\psi}{dx} \end{aligned}$$

Thus $G = P$, and **momentum is a generator of translations**, just like classical mechanics. The explicit form of $T(\varepsilon)$ is

$$T(\varepsilon) = I - \frac{i\varepsilon}{\hbar} P$$

We can also find a quantum analogue of the conservation law. Note that $\langle H \rangle$ must remain the same, so

$$\begin{aligned} \langle \psi|H|\psi\rangle &= \langle \psi|T^\dagger(\varepsilon)HT(\varepsilon)|\psi\rangle \\ &= \langle \psi|\left(I + \frac{i\varepsilon}{\hbar}P\right)H\left(I - \frac{i\varepsilon}{\hbar}P\right)|\psi\rangle \\ &= \langle \psi|H|\psi\rangle + \frac{i\varepsilon}{\hbar} \langle \psi|[P, H]|\psi\rangle + O(\varepsilon^2) \end{aligned}$$

Thus $\langle [P, H] \rangle = 0$, and by Ehrenfest's theorem, P is a time-independent variable when averaged, i.e.

$$\langle \dot{P} \rangle = 0$$

In general, the infinitesimal canonical transformation generated by $g(x, p)$, $\delta\omega = \varepsilon\{\omega, g\}$, corresponds to the infinitesimal unitary transformation $U_G(\varepsilon) = I - i\varepsilon/\hbar G$.

$$\delta\Omega = -\frac{i\varepsilon}{\hbar}[\Omega, G]$$

9.1.3 Finite Translations

Divide interval of length a into N segments, then take the limit as $N \rightarrow \infty$.

$$T(a/N) = I - \frac{ia}{\hbar N}P$$

$$T(a) = \lim_{N \rightarrow \infty} \left[I - \frac{ia}{\hbar N}P \right]^N = e^{-iaP/\hbar}$$

In the X basis, $P = -i\hbar \frac{d}{dx}$, so

$$T(a) = e^{-a d/dx} = 1 - a \frac{d}{dx} + \frac{a^2}{2!} \frac{d^2}{dx^2} + \frac{a^3}{3!} \frac{d^3}{dx^3} - \dots$$

$$\langle x|T(a)|\psi\rangle = \psi(x) - a \frac{d\psi}{dx} + \frac{a^2}{2!} \frac{d^2\psi}{dx^2} - \dots = \psi(x - a)$$

9.2 Invariance Under Transformations

9.2.1 Time-Translation Invariance

Prepare two systems at times t_1 and t_2 , both in state $|\psi\rangle$. Evolve under $U = e^{-iHt/\hbar}$.

$$|\psi(t_1 + \varepsilon)\rangle \approx \left[I - \frac{i\varepsilon}{\hbar} H(t_1) \right] |\psi_0\rangle$$

$$|\psi(t_2 + \varepsilon)\rangle \approx \left[I - \frac{i\varepsilon}{\hbar} H(t_2) \right] |\psi_0\rangle$$

The outcome is the same if $H(t_2) |\psi_0\rangle = H(t_1) |\psi_0\rangle$. Since ψ_0 and t_1, t_2 are arbitrary, $dH/dt = 0$, i.e. the system has no t dependence.

By Ehrenfest's theorem, $i\hbar \frac{d}{dt} \langle \Omega \rangle = \langle [\Omega, H] \rangle$. With $\Omega = H$,

$$\langle \dot{H} \rangle = 0$$

Which is the law of conservation of energy. Experiments that conserve energy have the same outcome regardless of where it happens.

9.2.2 Parity Invariance

Define the discrete parity operator Π such that $\Pi |x\rangle = |-x\rangle$ and $\Pi |p\rangle = |-p\rangle$. In the x basis, action of the Π on an arbitrary $|\psi\rangle$ is

$$\begin{aligned}\Pi |\psi\rangle &= \Pi \int_{-\infty}^{\infty} |x\rangle \langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty} |-x\rangle \langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty} |x'\rangle \langle -x'|\psi\rangle dx \\ \langle x|\Pi|\psi\rangle &= \psi(-x)\end{aligned}$$

Note that $\Pi^2 = I$. Thus Π has eigenvalues ± 1 , $\Pi = \Pi^{-1}$, and Π is both Hermitian and unitary.

H is parity-invariant if 2 systems that differ by a parity, after evolution, also differ by a parity. I.e.

$$\Pi^\dagger H(X, P)\Pi = H(-X, -P) = H(X, P)$$

Thus $[\Pi, U(t)] = 0$ as well.

All natural laws are invariant under spatial translations (homogeneity of space), temporal translations (homogeneity of time), and rotations (isotropic). But the weak interaction is **not invariant under parity**. The mirror-reflected scenario does not reflect a possible physics.

9.2.3 Time-Reversal Symmetry

Let the system start in (x_0, p_0) and come to (x, p) after time T . Reverse the momentum to get $(x, -p)$, then let the system evolve for time T . If we get $(x_0, -p_0)$, the system is time-reversal invariant (TIR). Note that the **momentum is reversed**.

TIR is not possible if e.g. the potential/force depends on t or x . For a particle with time-independent H , the complex conjugate performs the momentum-reversal.

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi(x, t)$$

For TIR to hold, start at $\psi(x, 0)$, evolve for T , complex conjugate, evolve for T , and we should get $\psi^*(x, 0)$.

$$\psi(x, 0) \rightarrow e^{-iHT/\hbar} \psi(x, T) \rightarrow e^{iH^*T/\hbar} \psi^*(x, T) \rightarrow e^{-iHT/\hbar} e^{-iH^*T/\hbar} \psi^*(x, 0)$$

So TIR holds iff. $H = H^*$, i.e. if H is real.

9.3 Rotations

In 2D classical space,

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} p_x \\ p_y \end{pmatrix} \rightarrow \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix}$$

Now consider an infinitesimal rotation $\varepsilon_z \hat{k}$ around the z -axis. Let L_z be the generator of infinitesimal rotations. The rotation operator is then

$$U[R(\varepsilon_z \hat{k})] = I - \frac{i\varepsilon}{\hbar} L_z$$

From the classical case,

$$U[R] |x, y\rangle = |x \cos \phi - y \sin \phi, x \sin \phi + y \cos \phi\rangle \approx |x - y\varepsilon_z, x\varepsilon_z + y\rangle$$

In the quantum case,

$$\begin{aligned} U[R] |\psi\rangle &= U[R] \int |x, y\rangle \langle x, y|\psi\rangle dx dy \\ &= \int |x - y\varepsilon, x\varepsilon + y\rangle \langle x, y|\psi\rangle dx dy \\ &= \int |x', y'\rangle \langle x' + y\varepsilon, y' - x\varepsilon|\psi\rangle dx' dy' \\ \langle x, y|U[R]|\psi\rangle &= \psi(x + y\varepsilon_z, y - x\varepsilon_z) \end{aligned}$$

With $U[R] = I - i\varepsilon_z L_z/\hbar$, we can see that

$$L_z \rightarrow x \left(-i\hbar \frac{\partial}{\partial y} \right) - y \left(-i\hbar \frac{\partial}{\partial x} \right) \equiv X P_y - Y P_x$$

The operator has an easier form in polar coordinates:

$$\begin{aligned} L_z &\rightarrow -i\hbar \frac{\partial}{\partial \phi} \\ U[R(\phi_0)] &\rightarrow \exp \left(-\phi_0 \frac{\partial}{\partial \phi} \right) \end{aligned}$$

Call L_z the **angular momentum operator**, which generates infinitesimal rotations about the z -axis. If the system is rotationally invariant, the experiment and its rotated version must give the same outcome.

$$\begin{aligned} U^\dagger[R] H(X, P_x; Y, P_y) U[R] &= H(X, P_x; Y, P_y) \\ [L_z, H] &= 0 \longrightarrow L_z \text{ conserved} \end{aligned}$$

The equation $[L_z, H] = 0$ imply that L_z and H have a simultaneous eigenbasis. The eigenvectors of L_z are given by $L_z |l_z\rangle = l_z |l_z\rangle$. In polar coordinates,

$$\begin{aligned} -i\hbar \frac{\partial \psi_{l_z}}{\partial \phi} &= l_z \psi_{l_z}(\rho, \phi) \\ \psi_{l_z}(\rho, \phi) &= R(\rho) e^{il_z \phi/\hbar} \end{aligned}$$

By forcing Hermiticity, $\langle \psi_1 | L_z | \psi_2 \rangle = \langle \psi_2 | L_z | \psi_1 \rangle^*$, we get $\psi(\rho, 0) = \psi(\rho, 2\pi)$. Thus

$$l_z = m\hbar \quad , \quad m = 0, \pm 1, \pm 2, \dots$$

m is the **magnetic quantum number**.

9.4 Summary: J, L, S

Operator	Measurement outcome	Definition
L^2 L_z	$\hbar^2 l(l+1)$, $l = 0, 1, 2, \dots$ $\hbar m_l$, $m_l = -l, -l+1, \dots, l-1, l$	l = orbital/azimuthal quantum number m_l = magnetic quantum number
S^2 S_z	$\hbar^2 s(s+1)$, $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ $\hbar m_s$, $m_s = -s, -s+1, \dots, s-1, s$	s is spin m_s = spin projection quantum number
J^2 J_z	$\hbar^2 j(j+1)$, $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ $\hbar m_j$	j = total angular momentum quantum number m_j = total angular mom. projection quantum number

10 Angular Momentum

10.1 Angular Momentum in 3 Dimensions

We follow the previous analysis and extend to 3 dimensions:

$$\begin{aligned} L_X &= YP_Z - ZP_Y \\ L_Y &= ZP_X - XP_Z \\ L_Z &= XP_Y - YP_X \end{aligned}$$

These are the generator of infinitesimal rotations:

$$U[R(\vec{\theta})] = \lim_{N \rightarrow \infty} \left(I - \frac{i}{\hbar} \frac{\theta}{N} \hat{\theta} \cdot \mathbf{L} \right)^N = e^{-i\vec{\theta} \cdot \mathbf{L}/\hbar}$$

Furthermore, the angular momentum operators satisfy

$$\mathbf{L} \times \mathbf{L} = i\hbar \mathbf{L} \quad \left\{ \begin{aligned} [L_x, L_y] &= i\hbar L_z \\ [L_y, L_z] &= i\hbar L_x \\ [L_z, L_x] &= i\hbar L_y \end{aligned} \right.$$

The total angular momentum operator $L^2 = L_x^2 + L_y^2 + L_z^2$ satisfies $[L^2, L_i] = 0$. Thus we can find a simultaneous eigenbasis for L^2 and L_i . Furthermore, if H is rotationally invariant under arbitrary $U[R]$, then there exists a simultaneous eigenbasis for H, L^2, L_i .

$$U^\dagger[R] H U[R] = H \quad \longrightarrow \quad [H, L_i] = 0 \quad \text{and} \quad [H, L^2] = 0$$

However, we can't find a simultaneous eigenbasis for all L_x, L_y, L_z because they don't commute.

In spherical coordinates,

$$\begin{aligned} L_z &\rightarrow -i\hbar \frac{\partial}{\partial \phi} \\ L^2 &\rightarrow -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \end{aligned}$$

10.2 Relation between L^2 and L_z

Let $|\alpha, \beta\rangle$ be the common eigenbasis of L^2 and L_z .

$$\begin{aligned} L^2 |\alpha, \beta\rangle &= \alpha |\alpha, \beta\rangle \\ L_z |\alpha, \beta\rangle &= \beta |\alpha, \beta\rangle \end{aligned}$$

Define the lowering/raising operators $L_{\pm} = L_x \pm iL_y$ such that $[L^2, L_{\pm}] = 0$ and $[L_z, L_{\pm}] = \pm\hbar L_{\pm}$. The operators $L_{\pm}$ raise/lower the eigenvalue of L_z by $\hbar$. E.g.:

$$\begin{aligned} L_z L_+ |\alpha, \beta\rangle &= (\hbar L_+ + L_+ L_z) |\alpha, \beta\rangle = (\beta + \hbar) L_+ |\alpha, \beta\rangle \\ L_+ |\alpha, \beta\rangle &= C_+(\alpha, \beta) |\alpha, \beta + \hbar\rangle \\ L_- |\alpha, \beta\rangle &= C_-(\alpha, \beta) |\alpha, \beta - \hbar\rangle \end{aligned}$$

But there must be some bounds in place, since $|l_z| \leq \sqrt{l^2}$. In QM, $\langle \alpha\beta | L^2 - L_z^2 | \alpha\beta \rangle \geq 0$, so $\alpha \geq \beta^2$. Then there must exist a state $|\alpha\beta_{\max}\rangle$ where β can no longer be raised. From $L_+ |\alpha\beta_{\max}\rangle = 0$ and $L_- |\alpha\beta_{\min}\rangle = 0$,

$$\begin{aligned} \alpha &= \beta_{\max}(\beta_{\max} + \hbar) \\ \alpha &= \beta_{\min}(\beta_{\min} - \hbar) \\ \beta_{\min} &= -\beta_{\max} \end{aligned}$$

Say we go from $\beta_{\min}$ to $\beta_{\max}$ in k steps of $\hbar$. Then

$$\begin{aligned} \beta_{\max} - \beta_{\min} &= 2\beta_{\max} = \hbar k \\ \beta_{\max}/2 &= k/2, \quad k = 0, 1, 2, \dots \\ \alpha &= \hbar^2 \left(\frac{k}{2}\right) \left(\frac{k}{2} + 1\right) \end{aligned}$$

We call $k/2$ the **angular momentum of the state**.

10.3 Total Angular Momentum: $J = L + S$

When rotating an arbitrary state vector ψ with L_z , we 1) reassign (x, y) to (x', y') , 2) rotate vector itself by an infinitesimal angle.

- L_z does part 1.
- S , the **spin matrix**, does part 2.

Thus the total generator of infinitesimal rotations is

$$\begin{aligned} J_i &= L_i + S_i \\ \mathbf{J} \times \mathbf{J} &= i\hbar \mathbf{J} \end{aligned}$$

The eigenvalue of J^2 and J_z , following previous analysis, is

$$\begin{aligned} J^2 |jm\rangle &= j(j+1)\hbar^2 |jm\rangle, \quad j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots \\ J_z |jm\rangle &= m\hbar |jm\rangle, \quad m = j, j-1, \dots, -j+1, -j \end{aligned}$$

The raising and lowering operators (to appropriate normalization) can be defined:

$$\begin{aligned} J_{\pm} &= J_x \pm iJ_y \\ J_+ |jm\rangle &= \hbar\sqrt{(j-m)(j+m+1)} |j, m+1\rangle \\ J_- |jm\rangle &= \hbar\sqrt{(j+m)(j-m+1)} |j, m-1\rangle \end{aligned}$$

Since acting on $|j, \pm j\rangle$ results in $|0\rangle$, for each state with total angular momentum (J^2) equal to j , there's only $(2j+1)$ distinct states: one for each J_z eigenvalue $j\hbar, (j-1)\hbar, \dots$

Assume $J = L$ to disregard the spin matrix. Simultaneous eigenfunctions to H, L_z, L^2 have the form

$$\begin{aligned} \psi_{Elm}(r, \theta, \phi) &= R_{Elm}(r) Y_l^m(\theta, \phi) \\ Y_l^m(\theta, \phi) &= \text{Spherical harmonics} \end{aligned}$$

10.4 Case Study: Free Particle in Spherical Coordinates

10.4.1 Schrodinger Equation in Spherical Coordinates

We start with the Schrodinger equation in spherical coordinates:

$$\begin{aligned} \left[-\frac{\hbar^2}{2\mu} \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) + V(r) \right] \psi_E &= E\psi_E \\ \psi_{Elm}(r, \theta, \phi) &= R_{Elm}(r) Y_l^m(\theta, \phi) \end{aligned}$$

We note that the terms in ∇^2 look similar to the spherical-coordinate form of L^2 . After substituting,

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \right) + V(r) \right] R(r) = ER(r)$$

We introduce $U_{El} = R_{El}/r$. This simplifies the equation to

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] U_{El} = EU_{El}$$

This is just like the normal Schrodinger equation, except the range is $0 \leq r \leq \infty$ and there's a repulsive centrifugal force $l(l+1)\hbar^2/2\mu r^2$. To prevent blowups at $r \rightarrow 0$, we enforce $U_{El} \rightarrow 0$ as $r \rightarrow 0$. To make U_{El} in the Hilbert space, we enforce $U_{El} \rightarrow 0$ or $U_{El} \rightarrow e^{ikr}$ as $r \rightarrow \infty$.

If $V(r)$ is less singular than $1/r^2$, then $U_{El}'' \simeq \frac{l(l+1)}{r^2} U_{El}$. If $U_{El} \sim r^\alpha$, then

$$U_{El} \sim \begin{cases} r^{l+1} & \text{regular} \\ r^{-l} & \text{irregular} \end{cases}$$

10.4.2 Free Particle in Spherical Coordinates

We start with the Schrodinger equation in spherical coordinates, introducing $U_{El} = R_{El}/r$ as before. We get

$$\left[\frac{d^2}{dr^2} + k^2 - \frac{l(l+1)}{r^2} \right] U_{El} = 0, \quad k^2 = \frac{2\mu E}{\hbar^2}$$

$$\left[-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} \right] U_l = U_l, \quad \rho = kr$$

Define the raising and lowering operators $d_l^\dagger$ and d_l such that

$$(d_l d_l^\dagger) U_l = U_l \quad \longrightarrow \quad \begin{cases} d_l = \frac{d}{d\rho} + \frac{l+1}{\rho} \\ d_l^\dagger = -\frac{d}{d\rho} + \frac{l+1}{\rho} \end{cases}$$

We have $d_l^\dagger d_l d_l^\dagger U_l = d_l^\dagger U_l$, or $d_{l+1}^\dagger d_{l+1} (d_l^\dagger U_l) = (d_l^\dagger U_l)$. Thus we can view $d_l^\dagger U_l \sim U_{l+1}$, hence the name “raising” operator. With $d_l^\dagger$ and d_l defined, we can build a ladder of solutions from $l = 0$.

At $l = 0$, two solutions are possible from the differential equation:

$$\begin{aligned} 1) U_0^A &= \sin \rho & \longrightarrow & R_0^A = \frac{\sin \rho}{\rho} \\ 2) U_0^B &= -\cos \rho & \longrightarrow & R_0^B = \frac{-\cos \rho}{\rho} \end{aligned}$$

10.4.3 Bessel and Neumann Wavefunctions

We seek to find a relationship between R_l and R_0 . Note that

$$\begin{aligned} \rho R_{l+1} &= d_l^\dagger (\rho R_l) = \left[-\frac{d}{d\rho} + \frac{l+1}{\rho} \right] (\rho R_l) \\ R_{l+1} &= \left[-\frac{d}{d\rho} + \frac{l}{\rho} \right] R_l = \rho^l \left(-\frac{d}{d\rho} \right) \frac{R}{\rho^l} \\ \frac{R_{l+1}}{\rho_{l+1}} &= \left(-\frac{1}{\rho} \frac{d}{d\rho} \right)^2 \frac{R_l}{\rho^l} \\ &= \left(-\frac{1}{\rho} \frac{d}{d\rho} \right)^l \frac{R_0}{\rho^0} \end{aligned}$$

Thus:

$$R_l = (-\rho)^l \left(\frac{1}{\rho} \frac{\partial}{\partial \rho} \right)^l R_0$$

There are thus two families of solutions:

$$\text{Spherical Bessel functions of order } l: \quad R_l^A \equiv j_l(\rho) = (-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{\sin \rho}{\rho} \right)$$

$$\text{Spherical Neumann functions of order } l: \quad R_l^B \equiv n_l(\rho) = (-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{-\cos \rho}{\rho} \right)$$

As $\rho \rightarrow 0$,

$$\begin{aligned} j_l(\rho) &\rightarrow \frac{\rho^l}{(2l+1)!!} \\ n_l(\rho) &\rightarrow -\frac{(2l+1)!!}{\rho^{l+1}} \end{aligned}$$

These correspond to the two families of U_{El} described previously. As $\rho \rightarrow 0$, $n_l(\rho)$ is singular, so if we want a solution regular in all space,

$$\psi_{Elm}(r, \theta, \phi) = j_l(kr)Y_l^m(\theta, \phi), \quad E = \frac{\hbar^2 k^2}{2\mu}$$

10.5 The Isotropic Oscillator

The isotropic oscillator is just the 1d oscillator in 3d space:

$$\begin{aligned} H &= \frac{P_x^2 + P_y^2 + P_z^2}{2\mu} + \frac{1}{2}\mu\omega^2(X^2 + Y^2 + Z^2) \\ \psi_{Elm}(r, \theta, \phi) &= j_l(kr)Y_l^m(\theta, \phi) \end{aligned}$$

With the wavefunction for a spherically symmetric potential, we obtain the radial equation:

$$\left[\frac{dd^2}{dr^2} + \frac{2\mu}{\hbar^2} \left(E - \frac{1}{2}\mu\omega^2 r^2 - \frac{l(l+1)\hbar^2}{2\mu r^2} \right) \right] U_{El} = 0$$

At $r \rightarrow \infty$, we have $U \sim e^{-y^2/2}$, where $y = r\sqrt{\mu\omega/\hbar}$. This is dimensionless, so let $U(y) = e^{-y^2/2}v(y)$. Furthermore, from previous analysis, we want $U(y) \rightarrow y^{l+1}$ as $r \rightarrow 0$. Together, we get the differential equation and a power series solution:

$$\begin{aligned} v'' - 2yv' + \left[2\lambda - 1 - \frac{l(l+1)}{y^2} \right] v &= 0, \quad \lambda = \frac{E}{\hbar\omega} \\ v(y) &= y^{l+1} \sum_0^\infty C_n y^n \end{aligned}$$

After messy solving, we get $E = \left(n + \frac{3}{2}\right) \hbar\omega$, where $n = 2k + 1$ is the **principal quantum number** with $k = 0, 1, 2, \dots$. The first few eigenstates are:

$n = 0$	$l = 0$	$m = 0$
$n = 1$	$l = 1$	$m = \pm 1, 0$
$n = 2$	$l = 0, 2$	$m = 0; m = \pm 2, \pm 1, 0$

10.6 The Hydrogen Atom

10.6.1 Schrodinger Equation

Since $m_p \gg m_e$, consider the electron to move around a spherically symmetric potential e/r^2 generated by a stationary proton.

The Schrodinger equation for spherical coordinates with $R_{El} = U_{El}/r$ is

$$\left[\frac{d^2}{dr^2} + \frac{2m}{\hbar^2} \left(E - \frac{e^2}{r} - \frac{l(l+1)\hbar^2}{2mr^2} \right) \right] U_{El} = 0$$

$$\psi_{Elm}(r, \theta, \phi) = R_{El}(r) Y_l^m(\theta, \phi)$$

- At $r \rightarrow \infty$, $d^2U/dr^2 \sim (-2mE/\hbar^2)U$. Let $\rho = \sqrt{2mW/\hbar^2}r$, where $W = -E$ is the binding energy. Then $U_{El} \rightarrow e^{-\rho}$ as $r \rightarrow \infty$.
- At $r \rightarrow 0$, we want $U \sim r^{l+1}$ as described previously.

Let $U_{El} = e^{-\rho}v_{El}$ and plug in to the Schrodinger equation. We get

$$\frac{d^2v}{d\rho^2} - 2\frac{dv}{d\rho} + \left[\frac{e^2\lambda}{\rho} - \frac{l(l+1)}{\rho^2} \right] v = 0, \quad \lambda = \sqrt{\frac{2m}{\hbar^2}W}$$

We try the ansatz $v = \rho^{l+1} \sum C_k \rho^k$ and after a lot of messy algebra, get

$$\frac{C_{k+1}}{C_k} = \frac{-e^2\lambda + 2(k+l+1)}{(k+l+2)(k+l+1) - l(l+1)}$$

As $k \rightarrow \infty$, $C_{k+1}/C_k \rightarrow 2/k$, which accords with $v \sim \rho^m e^\rho$. This blows up as $\rho \rightarrow \infty$, so we need the series to cut off at some k , i.e. $2(k+l+1) = e^2\lambda$. Recalling the definition of lambda, we get

$$E = -W = -\frac{me^4}{2\hbar^2(k+l+1)^2}, \quad k, l = 0, 1, 2, \dots$$

Or with the **principal quantum number** $n = k + l + 1$,

$$E = -\frac{me^4}{2\hbar^2 n^2}, \quad n = 1, 2, 3, \dots$$

Spectroscopic notation: $l = 0, 1, 2, 3, \dots$ corresponds to $s, p, d, f, \dots$. Thus $1s$ means $(n = 1; l = 0)$, $2s$ means $(n = 2, l = 0)$, and $2p$ means $(n = 2; l = 1)$. m_l is clear from l .

A **Rydberg of energy** is $Ry = me^4/2\hbar^2$, thus $E_n = -Ry/n^2$ for $n = 1, 2, 3, \dots$